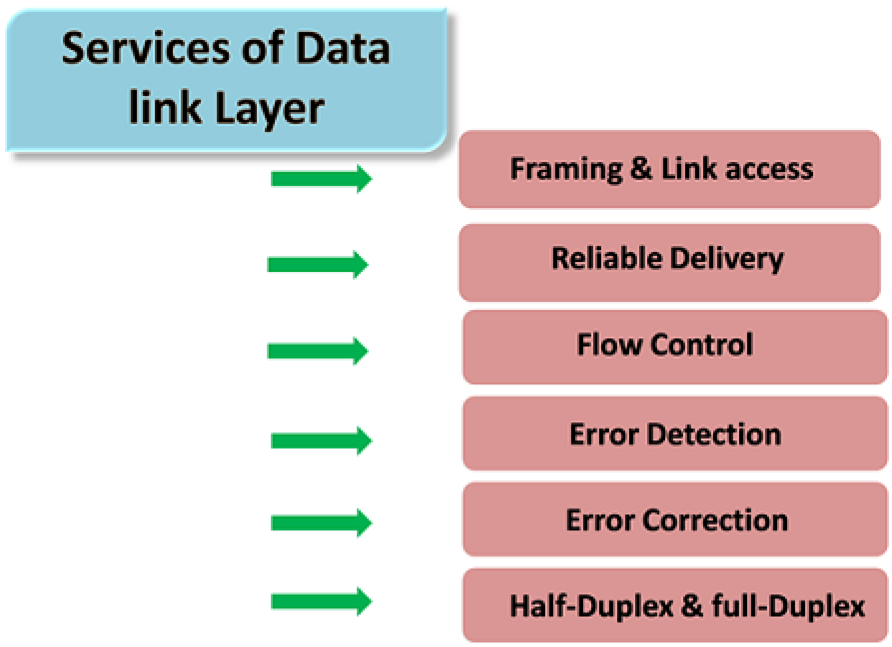
MODULE IV

- Link layer and Physical Layer
- Introduction to link layer
- Error detection (parity, checksum, and CRC)
- Multiple access protocols (collision and token based)
- IEEE 802.3 Ethernet
- Switching and bridging
- Media, Signal strength and interference
- Data encoding
- Ethernet switches, Routers MAC, ARP, FIB

<u>Data Link Layer</u>

- In the OSI model, the data link layer is a 4th layer from the top and 2nd layer from the bottom.
- The communication channel that connects the adjacent nodes is known as links, and in order to move the datagram from source to the destination, the datagram must be moved across an individual link.
- The main responsibility of the Data Link Layer is to transfer the datagram across an individual link.
- The Data link layer protocol defines the format of the packet exchanged across the nodes as well as the actions such as Error detection, retransmission, flow control, and random access.
- The Data Link Layer protocols are Ethernet, token ring, FDDI and PPP.
- An important characteristic of a Data Link Layer is that datagram can be handled by different link layer protocols on different links in a path. For example, the datagram is handled by Ethernet on the first link, PPP on the second link.

Following services are provided by the Data Link Layer:

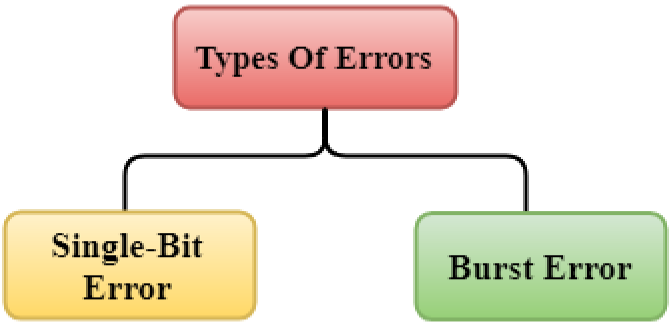


- **Framing & Link access:** Data Link Layer protocols encapsulate each network frame within a Link layer frame before the transmission across the link. A frame consists of a data field in which network layer datagram is inserted and a number of data fields. It specifies the structure of the frame as well as a channel access protocol by which frame is to be transmitted over the link.
- **Reliable delivery:** Data Link Layer provides a reliable delivery service, i.e., transmits the network layer datagram without any error. A reliable delivery service is accomplished with transmissions and acknowledgements. A data link layer mainly provides the reliable delivery service over the links as they have higher error rates and they can be corrected locally, link at which an error occurs rather than forcing to retransmit the data.
- **Flow control:** A receiving node can receive the frames at a faster rate than it can process the frame. Without flow control, the receiver's buffer can overflow, and frames can get lost. To overcome this problem, the data link layer uses the flow control to prevent the sending node on one side of the link from overwhelming the receiving node on another side of the link.
- **Error detection:** Errors can be introduced by signal attenuation and noise. Data Link Layer protocol provides a mechanism to detect one or more errors. This is achieved by adding error detection bits in the frame and then receiving node can perform an error check.
- **Error correction:** Error correction is similar to the Error detection, except that receiving node not only detect the errors but also determine where the errors have occurred in the frame.
- **Half-Duplex & Full-Duplex:** In a Full-Duplex mode, both the nodes can transmit the data at the same time. In a Half-Duplex mode, only one node can transmit the data at the same time.

Error Detection

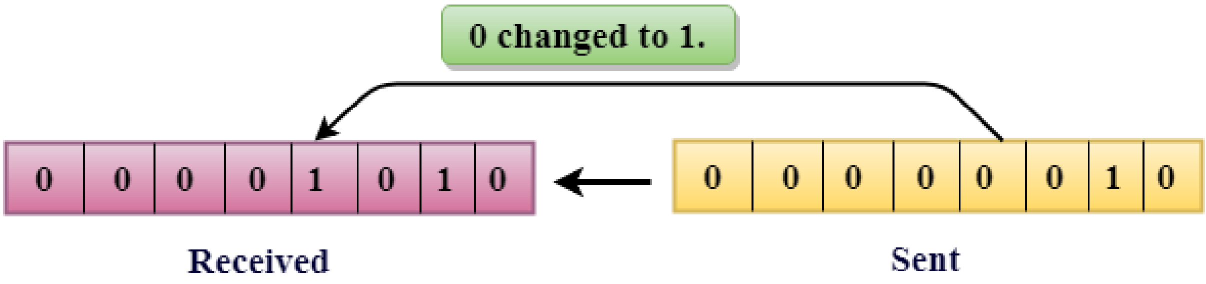
When data is transmitted from one device to another device, the system does not guarantee whether the data received by the device is identical to the data transmitted by another device. An Error is a situation when the message received at the receiver end is not identical to the message transmitted.

Types of Errors



- **Single-Bit Error:**

The only one bit of a given data unit is changed from 1 to 0 or from 0 to 1.



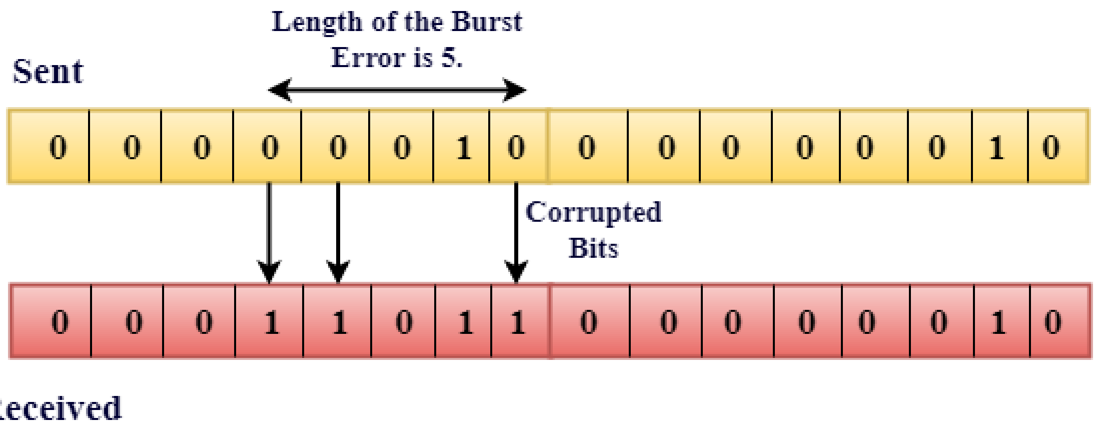
In the above figure, the message which is sent is corrupted as single-bit, i.e., 0 bit is changed to 1.

Single-Bit Error does not appear more likely in Serial Data Transmission. For example, Sender sends the data at 10 Mbps, this means that the bit lasts only for 1 s and for a single-bit error to occurred, a noise must be more than 1 s.

Single-Bit Error mainly occurs in Parallel Data Transmission. For example, if eight wires are used to send the eight bits of a byte, if one of the wires is noisy, then single-bit is corrupted per byte.

Burst Error:

The two or more bits are changed from 0 to 1 or from 1 to 0 is known as Burst Error. The Burst Error is determined from the first corrupted bit to the last corrupted bit.



The duration of noise in Burst Error is more than the duration of noise in Single-Bit. Burst Errors are most likely to occur in Serial Data Transmission. The number of affected bits depends on the duration of the noise and data rate.

Error Detecting Techniques:

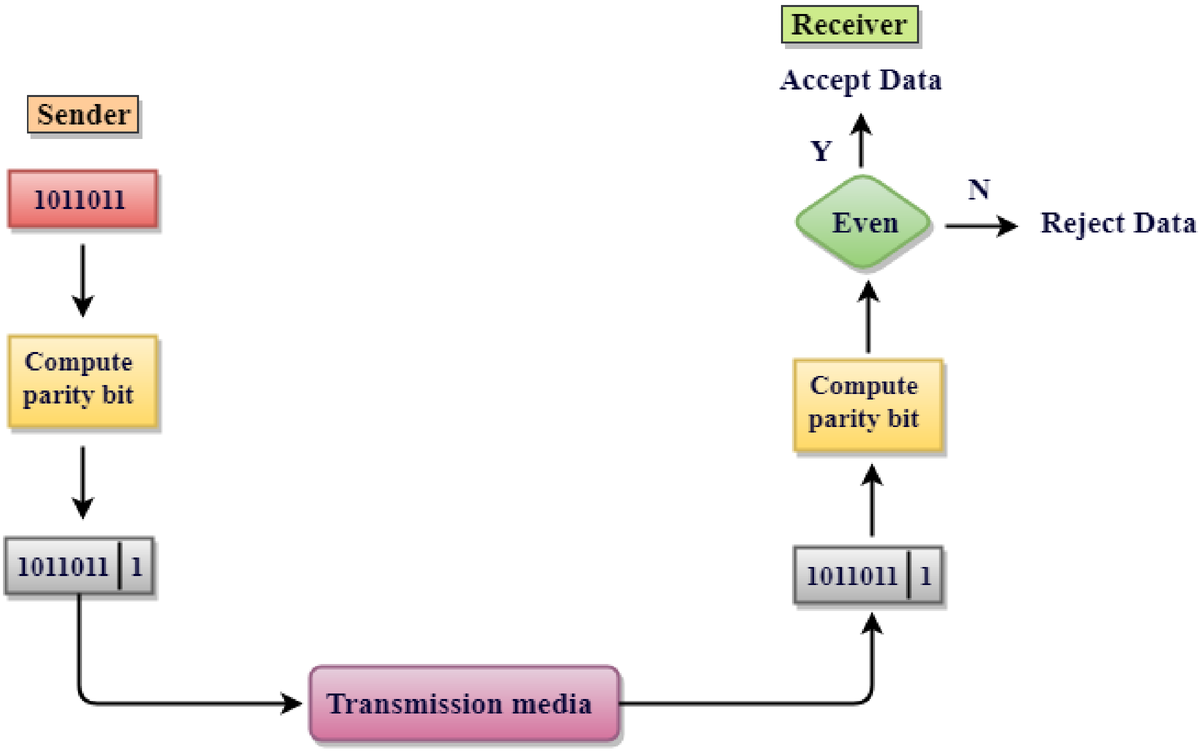
The most popular Error Detecting Techniques are:

- Single parity check
- Two-dimensional parity check
- Checksum
- Cyclic redundancy check

1.Single Parity Check

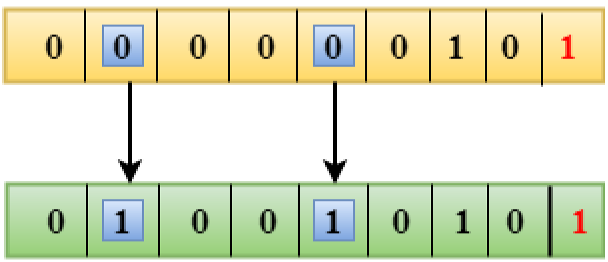
- Single Parity checking is the simple mechanism and inexpensive to detect the errors.
- In this technique, a redundant bit is also known as a parity bit which is appended at the end of the data unit so that the number of 1s becomes even. Therefore, the total number of transmitted bits would be 9 bits.

- If the number of 1s bits is odd, then parity bit 1 is appended and if the number of 1s bits is even, then parity bit 0 is appended at the end of the data unit.
- At the receiving end, the parity bit is calculated from the received data bits and compared with the received parity bit.
- This technique generates the total number of 1s even, so it is known as even-parity checking.



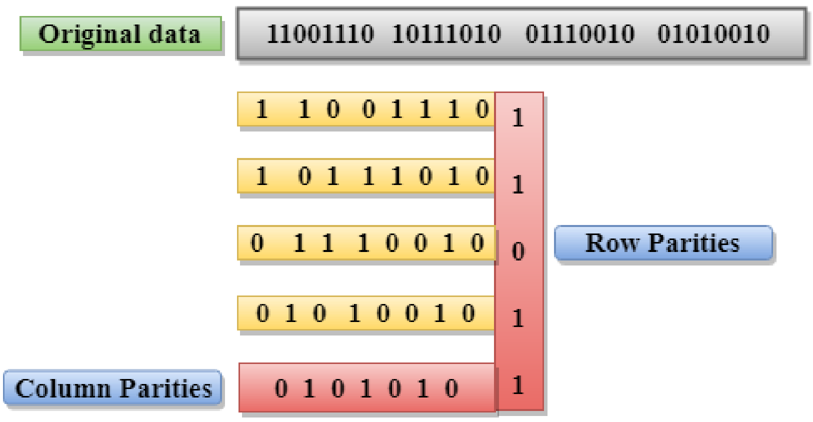
Drawbacks of Single Parity Checking

- It can only detect single-bit errors which are very rare.
- If two bits are interchanged, then it cannot detect the errors.



2.Two-Dimensional Parity Check

- Performance can be improved by using **Two-Dimensional Parity Check** which organizes the data in the form of a table.
- Parity check bits are computed for each row, which is equivalent to the single-parity check.
- In Two-Dimensional Parity check, a block of bits is divided into rows, and the redundant row of bits is added to the whole block.
- At the receiving end, the parity bits are compared with the parity bits computed from the received data.



Drawbacks Of 2D Parity Check

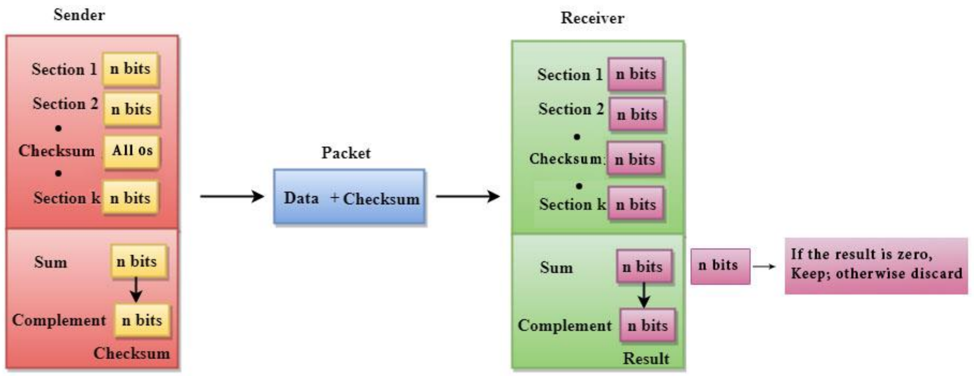
- If two bits in one data unit are corrupted and two bits exactly the same position in another data unit are also corrupted, then 2D Parity checker will not be able to detect the error.
- This technique cannot be used to detect the 4-bit errors or more in some cases.

3.Checksum

A Checksum is an error detection technique based on the concept of redundancy. **It is divided into two parts:**

- ### Checksum Generator

  A Checksum is generated at the sending side. Checksum generator subdivides the data into equal segments of n bits each, and all these segments are added together by using one's complement arithmetic. The sum is complemented and appended to the original data, known as checksum field. The extended data is transmitted across the network. Suppose L is the total sum of the data segments, then the checksum would be L.



1. The Sender follows the given steps:
2. The block unit is divided into k sections, and each of n bits.
3. All the k sections are added together by using one's complement to get the sum.
4. The sum is complemented and it becomes the checksum field.
5. The original data and checksum field are sent across the network.

- **Checksum Checker**

  A Checksum is verified at the receiving side. The receiver subdivides the incoming data into equal segments of n bits each, and all these segments are added together, and then this sum is complemented. If the complement of the sum is zero, then the data is accepted otherwise data is rejected.

  The Receiver follows the given steps:

1. The block unit is divided into k sections and each of n bits.
2. All the k sections are added together by using one's complement algorithm to get the sum.
3. The sum is complemented.
4. If the result of the sum is zero, then the data is accepted otherwise the data is discarded.

4.Cyclic Redundancy Check (CRC)

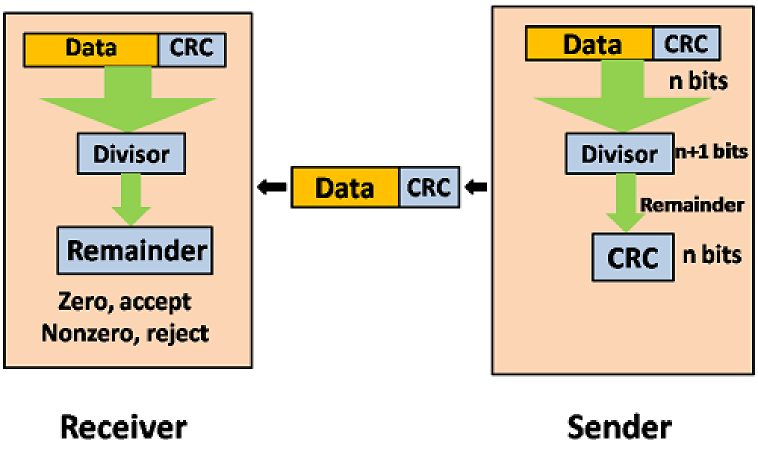
CRC is a redundancy error technique used to determine the error.

Following are the steps used in CRC for error detection:

- In CRC technique, a string of n 0s is appended to the data unit, and this n number is less than the number of bits in a predetermined number, known as division which is n+1 bit.
- Secondly, the newly extended data is divided by a divisor using a process is known as binary division. The remainder generated from this division is known as CRC remainder.
- Thirdly, the CRC remainder replaces the appended 0s at the end of the original data. This newly generated unit is sent to the receiver.
- The receiver receives the data followed by the CRC remainder. The receiver will treat this whole unit as a single unit, and it is divided by the same divisor that was used to find the CRC remainder.

If the resultant of this division is zero which means that it has no error, and the data is accepted.

If the resultant of this division is not zero which means that the data consists of an error. Therefore, the data is discarded.

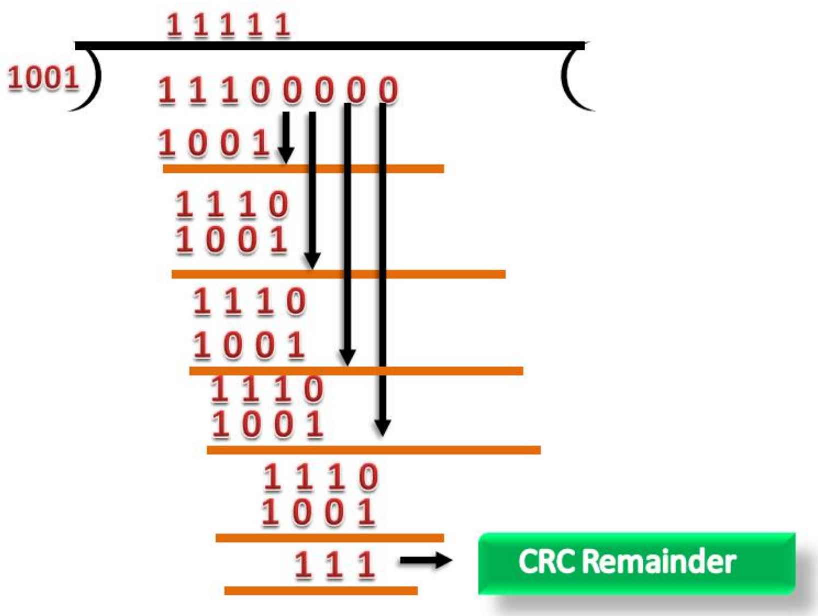


Let's understand this concept through an example:

Suppose the original data is 11100 and divisor is 1001.

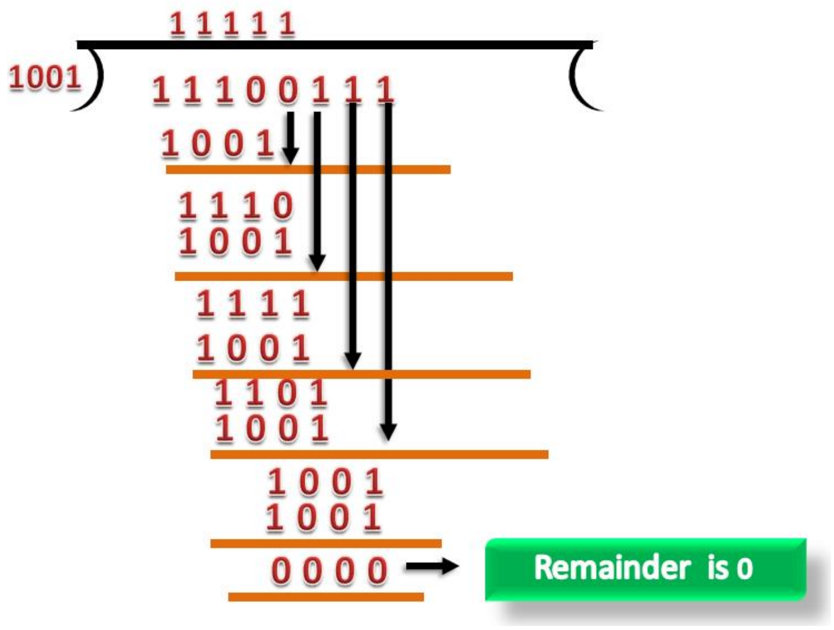
CRC Generator

- A CRC generator uses a modulo-2 division. Firstly, three zeroes are appended at the end of the data as the length of the divisor is 4 and we know that the length of the string 0s to be appended is always one less than the length of the divisor.
- Now, the string becomes 11100000, and the resultant string is divided by the divisor 1001.
- The remainder generated from the binary division is known as CRC remainder. The generated value of the CRC remainder is 111.
- CRC remainder replaces the appended string of 0s at the end of the data unit, and the final string would be 11100111 which is sent across the network.



CRC Checker

- The functionality of the CRC checker is similar to the CRC generator.
- When the string 11100111 is received at the receiving end, then CRC checker performs the modulo-2 division.
- A string is divided by the same divisor, i.e., 1001.
- In this case, CRC checker generates the remainder of zero. Therefore, the data is accepted.



Multiple Access Protocols-Collision and Token Based

The data link layer is used in a computer network to transmit the data between two devices or nodes. It divides the layer into parts such as **data link control** and the **multiple access resolution/protocol**. The upper layer has the responsibility to flow control and the error control in the data link layer, and hence it is termed as **logical of data link control**. Whereas the lower sub-layer is used to handle and reduce the collision or multiple access on a channel. Hence it is termed as media access control or the multiple access resolutions.

Data Link Control

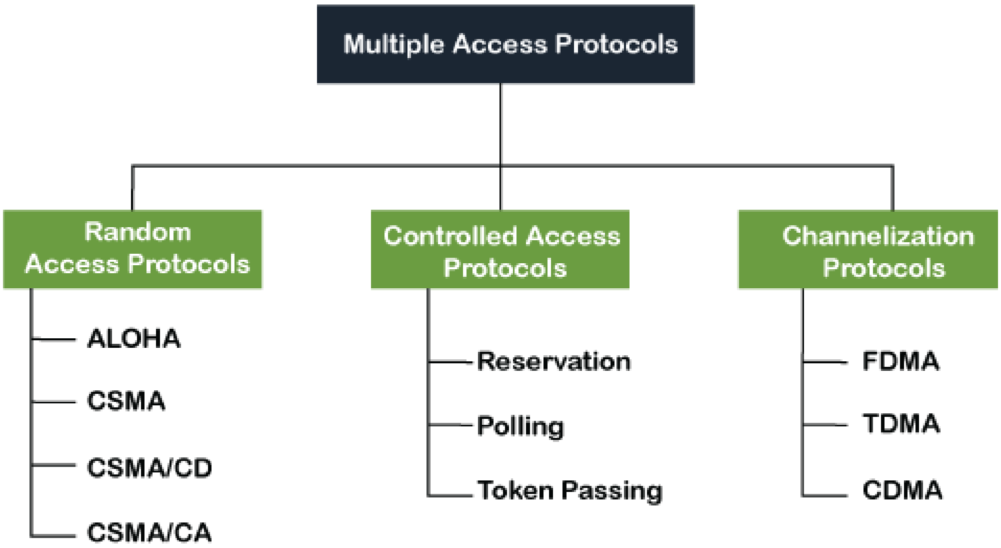
A data link control is a reliable channel for transmitting data over a dedicated link using various techniques such as framing, error control and flow control of data packets in the computer network.

What is a multiple access protocol?

When a sender and receiver have a dedicated link to transmit data packets, the data link control is enough to handle the channel. Suppose there is no dedicated path to communicate or transfer the data between two devices. In that case, multiple stations access the channel and simultaneously transmits the data over the channel. It may create collision and cross talk. Hence, the multiple access protocol is required to reduce the collision and avoid crosstalk between the channels.

For example, suppose that there is a classroom full of students. When a teacher asks a question, all the students (small channels) in the class start answering the question at the same time (transferring the data simultaneously). All the students respond at the same time due to which data is overlap or data lost. Therefore, it is the responsibility of a teacher (multiple access protocol) to manage the students and make them one answer.

Following are the types of multiple access protocol that is subdivided into the different process as:



A. Random Access Protocol

In this protocol, all the station has the equal priority to send the data over a channel. In random access protocol, one or more stations cannot depend on another station nor any station control another station. Depending on the channel's state (idle or busy), each station transmits the data frame. However, if more than one station sends the data over a channel, there may be a collision or data conflict. Due to the collision, the data frame packets may be lost or changed. And hence, it does not receive by the receiver end. Following are the different methods of random-access protocols for broadcasting frames on the channel.

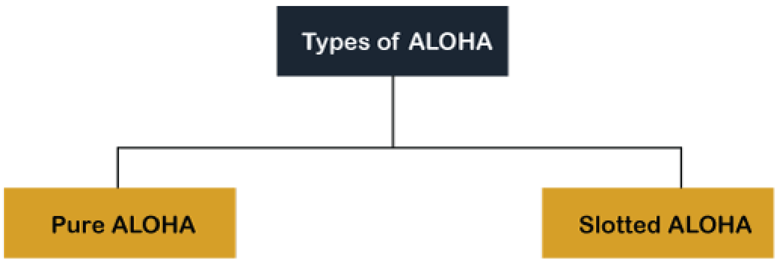
- Aloha
- CSMA
- CSMA/CD
- CSMA/CA

ALOHA Random Access Protocol

It is designed for wireless LAN (Local Area Network) but can also be used in a shared medium to transmit data. Using this method, any station can transmit data across a network simultaneously when a data frameset is available for transmission.

Aloha Rules

1. Any station can transmit data to a channel at any time.
2. It does not require any carrier sensing.
3. Collision and data frames may be lost during the transmission of data through multiple stations.
4. Acknowledgment of the frames exists in Aloha. Hence, there is no collision detection.
5. It requires retransmission of data after some random amount of time.



CSMA (Carrier Sense Multiple Access)

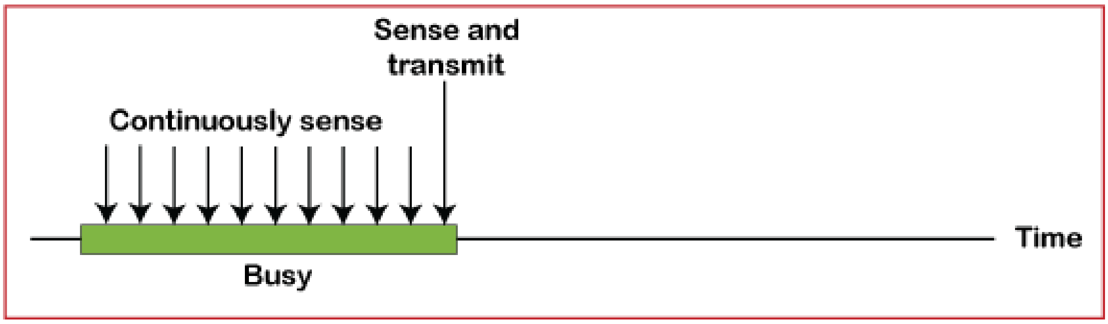
It is a **carrier sense multiple access** based on media access protocol to sense the traffic on a channel (idle or busy) before transmitting the data. It means that if the channel is idle, the station can send data to the channel. Otherwise, it must wait until the channel becomes idle. Hence, it reduces the chances of a collision on a transmission medium.

CSMA Access Modes

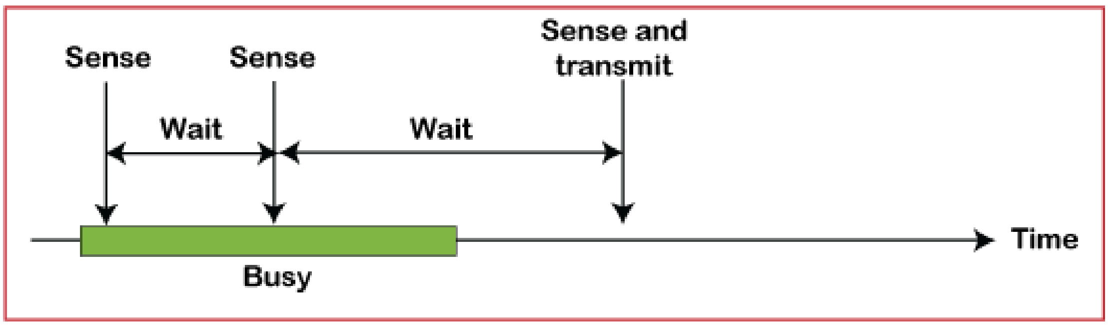
- **1-Persistent:** In the 1-Persistent mode of CSMA that defines each node, first sense the shared channel and if the channel is idle, it immediately sends the data. Else it must wait and keep track of the status of the channel to be idle and broadcast the frame unconditionally as soon as the channel is idle.
- **Non-Persistent:** It is the access mode of CSMA that defines before transmitting the data, each node must sense the channel, and if the

channel is inactive, it immediately sends the data. Otherwise, the station must wait for a random time (not continuously), and when the channel is found to be idle, it transmits the frames.

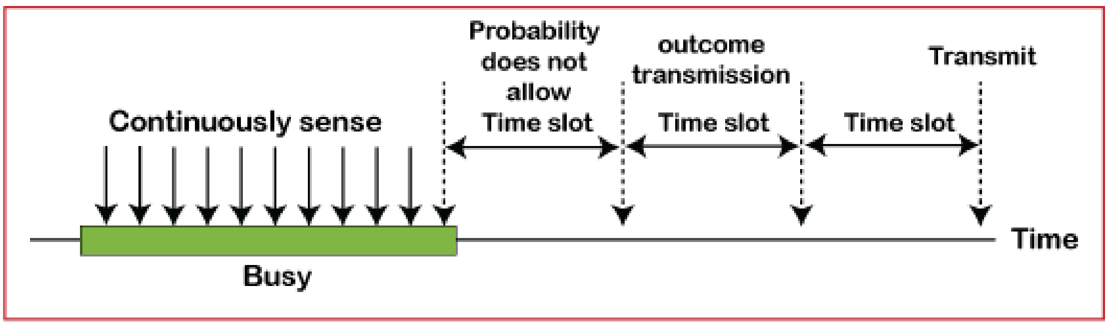
- **P-Persistent:** It is the combination of 1-Persistent and Non-persistent modes. The P-Persistent mode defines that each node senses the channel, and if the channel is inactive, it sends a frame with a **P** probability. If the data is not transmitted, it waits for a (**q = 1-p probability**) random time and resumes the frame with the next time slot.
- **O- Persistent:** It is an O-persistent method that defines the superiority of the station before the transmission of the frame on the shared channel. If it is found that the channel is inactive, each station waits for its turn to retransmit the data.



a. 1-persistent



b. Nonpersistent



c. p-persistent

CSMA/ CD

It is a **carrier sense multiple access/ collision detection** network protocol to transmit data frames. The CSMA/CD protocol works with a medium access control layer. Therefore, it first senses the shared channel before broadcasting the frames, and if the channel is idle, it transmits a frame to check whether the transmission was successful. If the frame is successfully received, the station sends another frame. If any collision is detected in the CSMA/CD, the station sends a jam/ stop signal to the shared channel to terminate data transmission. After that, it waits for a random time before sending a frame to a channel.

CSMA/ CA

It is a **carrier sense multiple access/collision avoidance** network protocol for carrier transmission of data frames. It is a protocol that works with a medium access control layer. When a data frame is sent to a channel, it receives an acknowledgment to check whether the channel is clear. If the station receives only a single (own) acknowledgments, that means the data frame has been successfully transmitted to the receiver. But if it gets two signals (its own and one more in which the collision of frames), a collision of the frame occurs in the shared channel. Detects the collision of the frame when a sender receives an acknowledgment signal.

Following are the methods used in the CSMA/CA to avoid the collision.

- **Interframe space**: In this method, the station waits for the channel to become idle, and if it gets the channel is idle, it does not immediately send the data. Instead of this, it waits for some time, and this time period is called the **Interframe** space or IFS. However, the IFS time is often used to define the priority of the station.
- **Contention window**: In the Contention window, the total time is divided into different slots. When the station/ sender is ready to transmit the data frame, it chooses a random slot number of slots as **wait time**. If the channel is still busy, it does not restart the entire process, except that it restarts the timer only to send data packets when the channel is inactive.
- **Acknowledgment**: In the acknowledgment method, the sender station sends the data frame to the shared channel if the acknowledgment is not received ahead of time.

B. Controlled Access Protocol

It is a method of reducing data frame collision on a shared channel. In the controlled access method, each station interacts and decides to send a data frame by a particular station approved by all other stations. It means that a single station cannot send the data frames unless all other stations are not approved. It has three types of controlled access: **Reservation, Polling**, and **Token Passing**.

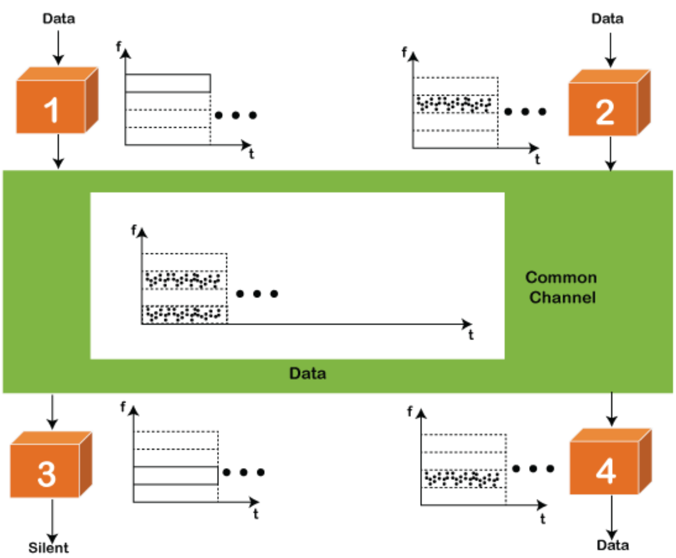
C. Channelization Protocols

It is a channelization protocol that allows the total usable bandwidth in a shared channel to be shared across multiple stations based on their time, distance and codes. It can access all the stations at the same time to send the data frames to the channel. Following are the various methods to access the channel based on their time, distance and codes:

1. FDMA (Frequency Division Multiple Access)
2. TDMA (Time Division Multiple Access)
3. CDMA (Code Division Multiple Access

FDMA

It is a frequency division multiple access (**FDMA**) method used to divide the available bandwidth into equal bands so that multiple users can send data through a different frequency to the subchannel. Each station is reserved with a particular band to prevent the crosstalk between the channels and interferences of stations.



TDMA

Time Division Multiple Access (**TDMA**) is a channel access method. It allows the same frequency bandwidth to be shared across multiple stations. And to avoid collisions in the shared channel, it divides the channel into different frequency slots that allocate stations to transmit the data frames. The same **frequency** bandwidth into the shared channel by dividing the signal into various time slots to transmit it. However, TDMA has an overhead of synchronization that specifies each station's time slot by adding synchronization bits to each slot.

CDMA

The Code Division Multiple Access is a channel access method. In CDMA, all stations can simultaneously send the data over the same channel. It means that it allows each station to transmit the data frames with full frequency on the shared channel at all times. It does not require the division of bandwidth on a shared channel based on time slots. If multiple stations send data to a channel simultaneously, their data frames are separated by a unique code sequence. Each station has a different unique code for transmitting the data over a shared channel. For example, there are multiple users in a room that are continuously speaking. Data is received by the users if only two-person interact with each other using the same language. Similarly, in the network, if different stations communicate with each other simultaneously with different code language.

IEEE 802.3 and Ethernet

Ethernet is a set of technologies and protocols that are used primarily in LANs. It was first standardized in 1980s by IEEE 802.3 standard. IEEE 802.3 defines the physical layer and the medium access control (MAC) sub-layer of the data link layer for wired Ethernet networks.

Ethernet is classified into two categories:

- classic Ethernet
- Switched Ethernet

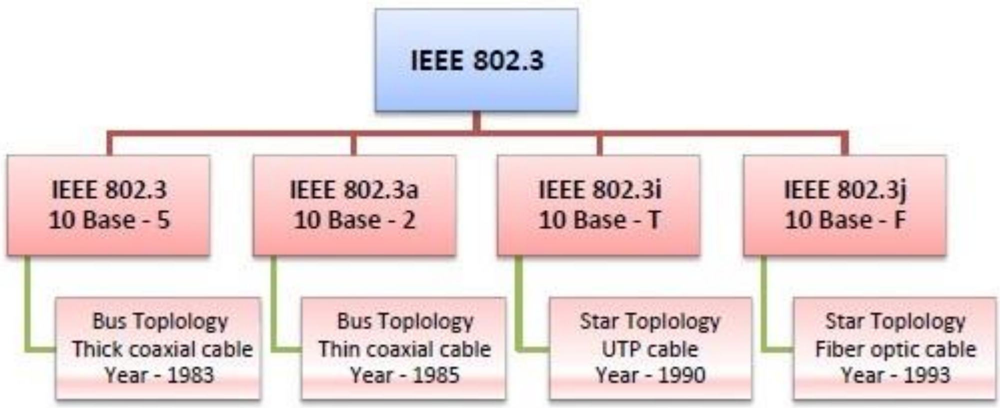
Classic Ethernet is the original form of Ethernet that provides data rates between 3 to 10 Mbps. The varieties are commonly referred as 10BASE-X. Here, 10 is the maximum throughput, i.e. 10 Mbps, BASE denoted use of baseband transmission, and X is the type of medium used. Most varieties of classic Ethernet have become obsolete in present communication scenario.

A switched Ethernet uses switches to connect to the stations in the LAN. It replaces the repeaters used in classic Ethernet and allows full bandwidth utilization.

IEEE 802.3 Popular Versions

There are a number of versions of IEEE 802.3 protocol. The most popular ones are -

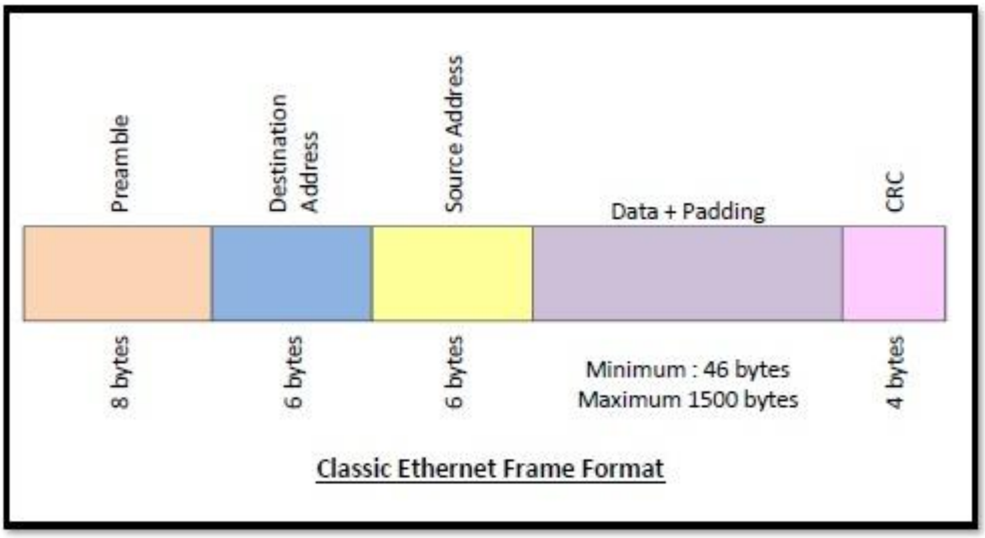
- **IEEE 802.3**: This was the original standard given for 10BASE-5. It used a thick single coaxial cable into which a connection can be tapped by drilling into the cable to the core. Here, 10 is the maximum throughput, i.e. 10 Mbps, BASE denoted use of baseband transmission, and 5 refers to the maximum segment length of 500m.
- **IEEE 802.3a**: This gave the standard for thin coax (10BASE-2), which is a thinner variety where the segments of coaxial cables are connected by BNC connectors. The 2 refers to the maximum segment length of about 200m (185m to be precise).
- **IEEE 802.3i**: This gave the standard for twisted pair (10BASE-T) that uses unshielded twisted pair (UTP) copper wires as physical layer medium. The further variations were given by IEEE 802.3u for 100BASE-TX, 100BASE-T4 and 100BASE-FX.
- **IEEE 802.3i**: This gave the standard for Ethernet over Fiber (10BASE-F) that uses fiber optic cables as medium of transmission.

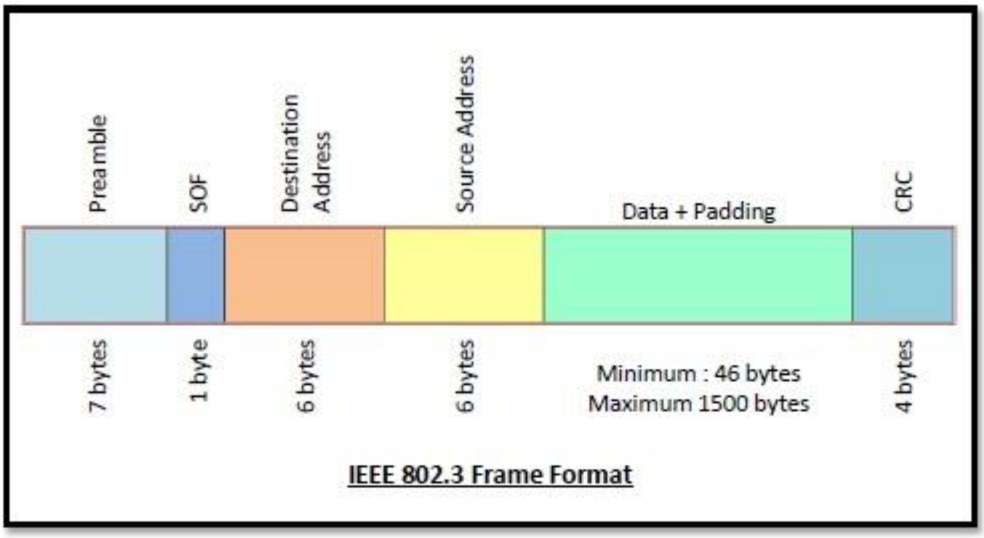


Frame Format of Classic Ethernet and IEEE 802.3

The main fields of a frame of classic Ethernet are –

- **Preamble**: It is the starting field that provides alert and timing pulse for transmission. In case of classic Ethernet, it is an 8-byte field and in case of IEEE 802.3 it is of 7 bytes.
- **Start of Frame Delimiter**: It is a 1-byte field in an IEEE 802.3 frame that contains an alternating pattern of ones and zeros ending with two ones.
- **Destination Address**: It is a 6-byte field containing physical address of destination stations.
- **Source Address**: It is a 6-byte field containing the physical address of the sending station.
- **Length**: It a 7 bytes field that stores the number of bytes in the data field.
- **Data**: This is a variable sized field carries the data from the upper layers. The maximum size of data field is 1500 bytes.
- **Padding**: This is added to the data to bring its length to the minimum requirement of 46 bytes.
- **CRC**: CRC stands for cyclic redundancy check. It contains the error detection information.





Switching

What is Switching?

Switching is the most valuable asset of computer networking. Every time in computer network you access the internet or another computer network outside your immediate location, or your messages are sent through a maze of transmission media and connection devices. The mechanism for exchange of information between different computer networks and network segments is called switching in Networking. On the other words we can say that any type signal or data element directing or Switching toward a particular hardware address or hardware pieces.

Hardware devices that can be used for switching or transferring data from one location to another that can use multiple layers of the Open Systems Interconnection (OSI) model. Hardware devices that can used for switching data in single location like collage lab is Hardware switch or hub but if you want to transfer data between to different location or remote location then we can use router or gateways.

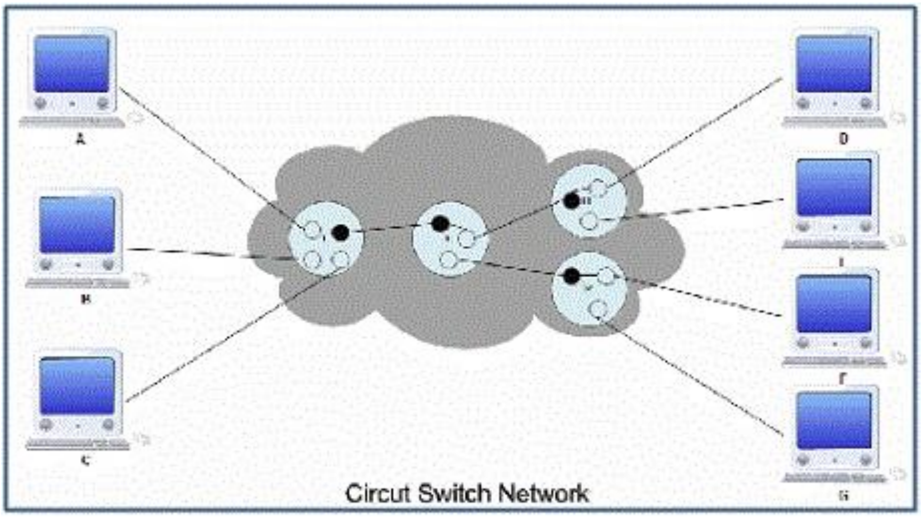
For example: whenever a telephone call is placed, there are numerous junctions in the communication path that perform this movement of data from one network onto another network. One of another example is gateway, that can be used by Internet Service Providers (ISP) to deliver a signal to another Internet Service Providers (ISP). For exchange of information between different locations various types of Switching Techniques are used in Networking.

Types of Switching Techniques

There are basically three types of switching methods are available.

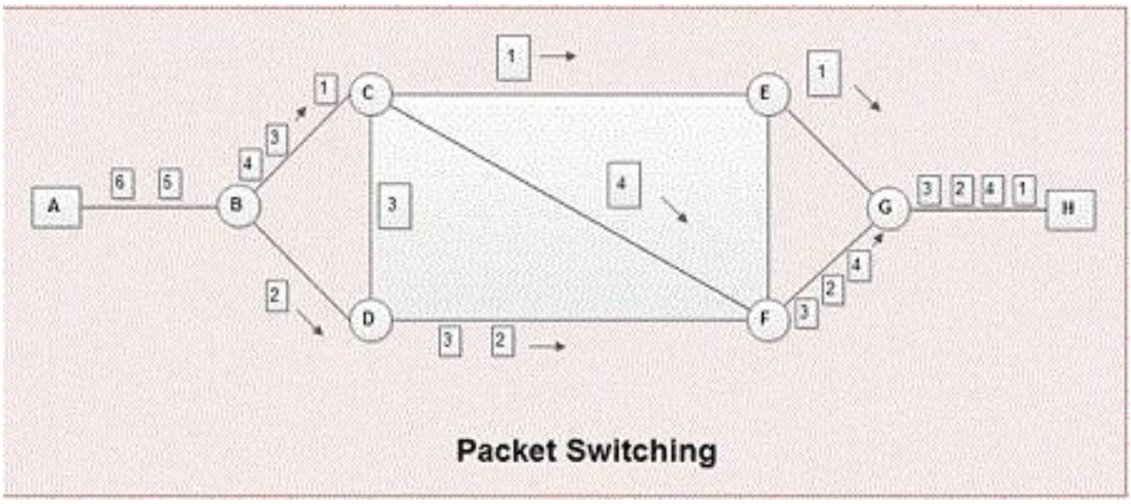
1.Circuit Switching

Circuit-switching is the real-time connection-oriented system. In Circuit Switching a dedicated channel (or circuit) is set up for a single connection between the sender and recipient during the communication session. In telephone communication system, the normal voice call is the example of Circuit Switching. The telephone service provider maintains an unbroken link for each telephone call. Circuit switching is pass through three phases, that are circuit establishment, data transfer and circuit disconnect



2.Packet Switching

The basic example of Packet Switching is the Internet. In Packet Switching, data can be fragmented into suitably-sized pieces in variable length or blocks that are called packets that can be routed independently by network devices based on the destination address contained certain “formatted” header within each packet. The packet switched networks allow sender and recipient without reserving the circuit. Multiple paths are existing between sender and recipient in a packet switching network. They does not require a call setup to transfer packets between sender and recipient.



Advantage Packet Switching

The main advantage of packet switching network is the efficiency. The other advantage is that, it is faults tolerant. In packet switching, the quality of data transmission is kept high (error free).

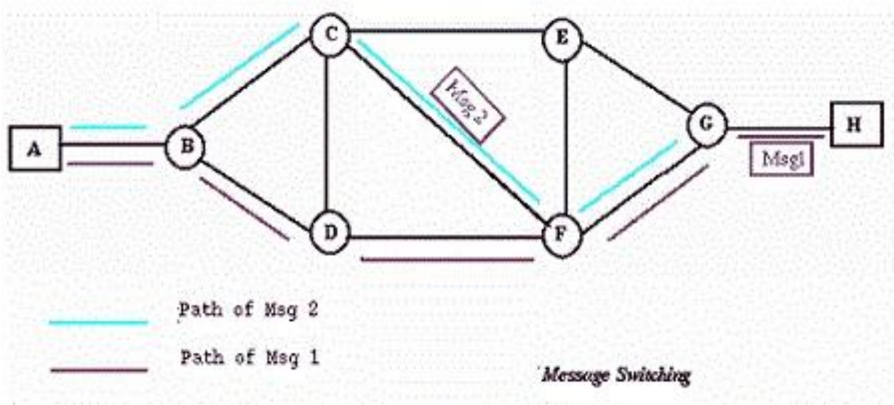
There are two type of Packet Switching connectionless or Connection-Oriented.

- Connectionless Packet Switching
  It is also known as datagram switching. In this type of network each packet routed individually by network devices based on the destination address contained within each packet. Due to each packet is routed individually, the result is that each packet is delivered out-of-order with different paths of transmission, it depend on the networking devices like (switches and routers) at any given time. After reaching recipient location, the packets are reassembling to the original form.
- Connection-Oriented Packet Switching

  It is also known as virtual circuit switching.in this type of Networking packets are send in sequential order over a defined route.

- Message Switching

  Message switching does not set up a dedicated channel (or circuit) between the sender and recipient during the communication session. In Message Switching each message is treated as an independent block. In this type of networking, each message is then transmitted from first network device to second network device through the internetwork i.e. message is transmitted from the sender to intermediator device.



The intermediate device stores the message for a time being, after inspects it for errors, intermediate device transmitting the message to the next node with its routing information. Because of this reason message switching networks are called store and forward networks in networking.

Bridging

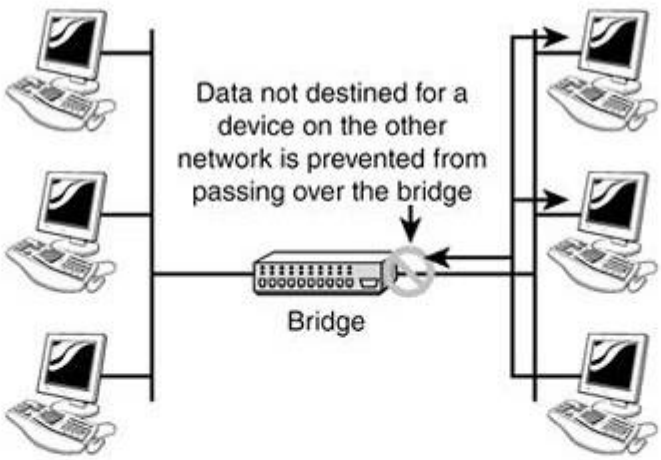
The computer network is the basis of communication in the field of IT (information technology). These networks are used in different ways by using different types of networks. The connection of a computer network can be done by connecting a set of computers so that the information can be shared. Historically, these networks are broken down into topologies, used for connecting computers. At present, the most commonly used topology is a collapsed ring because, this type of protocol supports the internet, LANs & WANs. Computer networks are used to share information by carrying a large number of tasks. Some of the functions of this network are communication with the help of an e-mail, messaging, video and device sharing like scanners, printers, sharing files, software's and permitting the users of the network to access and maintain the information easily. This article discusses an overview of a bridge in a computer network.

What is a Bridge in a Computer Network?

Definition: A bridge in a computer network is one kind of network device, used to separate a network into sections. Every section in the network represents a collision domain that has separate bandwidth. So that network performance can be improved using a bridge. In the OSI model, a bridge works at layer-2 namely the data link layer. The main function of this is to examine the incoming traffic and examine whether to filter it or forward it.

Working Principle

The working principle of a bridge is, it blocks or forwards the data depending on the destination MAC address and this address is written into every data frame.



In a computer network, a bridge separates a LAN into different segments like segment1 & segment2, etc and the MAC address of all the PCs can be stored into the table. For instance, PC1 transmits the data to PC2, where the data will transmit to the bridge first. So, the bridge reads the MAC address & decides whether to transmit the data to segment1 or segment2.

Therefore, the PC2 is accessible in segment1, which means the bridge transmits the data in segment1 only & eliminates all the connected PCs in segment2. In this way, the bridge reduces traffic in a computer network.

Use of Bridge in Computer Network

A bridge in a computer network connects with other bridge networks that utilize a similar protocol. These network devices work at the data link layer in an OSI model to connect two different networks and provide communication between them. Similar to hubs and repeaters, bridges broadcast data to each node. But, maintains the MAC (media access control) address table to find out new segments. So following transmissions are transmitted to the preferred receiver only.
A bridge utilizes a database to determine where to transmit otherwise remove the data frame.

Types of Bridges

Bridges in the computer network are classified into three types which include the following.

1.Transparent Bridge

As the name suggests, it is an invisible bridge in the computer network. The main function of this bridge is to block or forward the data depending on the MAC address. The other devices within the network are unaware of the existence of bridges. These types of bridges are most popular and operate in a transparent way to the entire networks which are connected to hosts.

This bridge saves the addresses of MAC within a table that is similar to a routing table. This estimates the information when a packet is routed to its position. So, it can also merge several bridges to check incoming traffic in a better way. These bridges are implemented mainly in Ethernet networks.

2.Translational Bridge

A translational bridge plays a key role in changing a networking system from one type to another. These bridges are used to connect two different networks like token ring & Ethernet. This bridge can add or remove the data based on the traveling direction, and forward the frames of the data link layer in between LANs which uses various types of network protocols. The different network connections are Ethernet to FDDI/token ring otherwise Ethernet on UTP (unshielded twisted pair) to coax & in between FOC and copper wiring.

3.Source-route Bridge

Source-route Bridge is one type of technique used for Token Ring networks and it is designed by IBM. In this bridge, the total frame route is embedded in one frame. So that it allows the bridge to make precise decisions of how the frame is forwarding using the network. By using this method, two similar network segments are connected to the data link layer. It can be done in a distributed way wherever end-stations join within the bridging algorithm.

Functions of Bridges in Network

The main functions of bridges in a computer network include the following.

This networking device is used for dividing local area networks into several segments. In the OSI model, it works under the data link layer. It is used to store the address of MAC in PC used in a network and also used for diminishing the network traffic.

Advantages/Disadvantages of Bridge

The advantages are

- It acts as a repeater to extend a network
- Network traffic on a segment can be reduced by subdividing it into network communications
- Collisions can be reduced.
- Some types of bridges connect the networks with the help of architectures & types of media.
- Bridges increase the available bandwidth to individual nodes because fewer network nodes share a collision domain
- It avoids waste BW (bandwidth)
- The length of the network can be increased.
- Connects different segments of network transmission

The disadvantages are

- It is unable to read specific IP addresses because they are more troubled with the MAC addresses.
- They cannot help while building the network between the different architectures of networks.
- It transfers all kinds of broadcast messages, so they are incapable to stop the scope of messages.

- These are expensive as we compare with repeaters
- It doesn't handle more variable & complex data load which occurs from WAN.

MEDIA

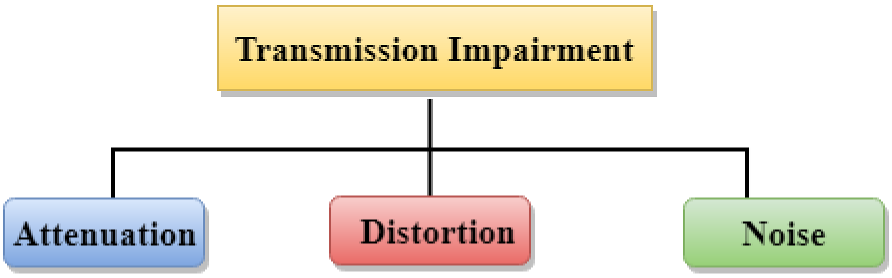
- Transmission media is a communication channel that carries the information from the sender to the receiver. Data is transmitted through the electromagnetic signals.
- The main functionality of the transmission media is to carry the information in the form of bits through **LAN** (Local Area Network).
- It is a physical path between transmitter and receiver in data communication.
- In a copper-based network, the bits in the form of electrical signals.
- In a fibre-based network, the bits in the form of light pulses.
- In **OSI** (Open System Interconnection) phase, transmission media supports the Layer 1. Therefore, it is considered to be as a Layer 1 component.
- The electrical signals can be sent through the copper wire, fibre optics, atmosphere, water, and vacuum.
- The characteristics and quality of data transmission are determined by the characteristics of medium and signal.
- Transmission media is of two types are wired media and wireless media. In wired media, medium characteristics are more important whereas, in wireless media, signal characteristics are more important.
- Different transmission media have different properties such as bandwidth, delay, cost and ease of installation and maintenance.
- The transmission media is available in the lowest layer of the OSI reference model, i.e., **Physical layer**.

Some factors need to be considered for designing the transmission media:

- **Bandwidth:** All the factors are remaining constant, the greater the bandwidth of a medium, the higher the data transmission rate of a signal.

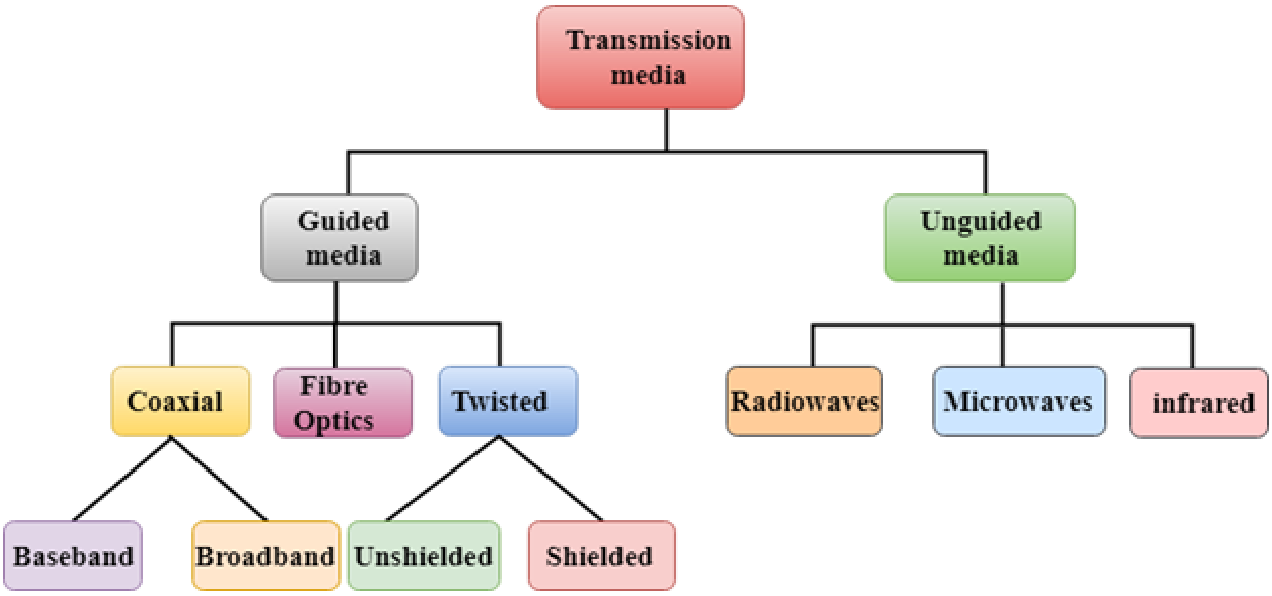
- **Transmission impairment:** When the received signal is not identical to the transmitted one due to the transmission impairment. The quality of the signals will get destroyed due to transmission impairment.
- **Interference:** An interference is defined as the process of disrupting a signal when it travels over a communication medium on the addition of some unwanted signal.

Causes of Transmission Impairment:



- **Attenuation:** Attenuation means the loss of energy, i.e., the strength of the signal decreases with increasing the distance which causes the loss of energy.
- **Distortion:** Distortion occurs when there is a change in the shape of the signal. This type of distortion is examined from different signals having different frequencies. Each frequency component has its own propagation speed, so they reach at a different time which leads to the delay distortion.
- **Noise:** When data is travelled over a transmission medium, some unwanted signal is added to it which creates the noise.

Classification of Transmission Media:



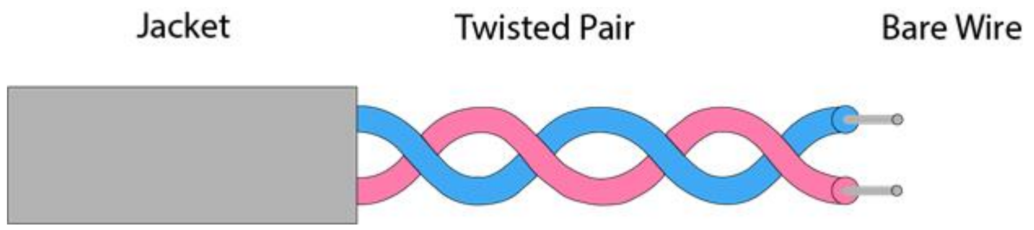
Guided Media

It is defined as the physical medium through which the signals are transmitted. It is also known as Bounded media.

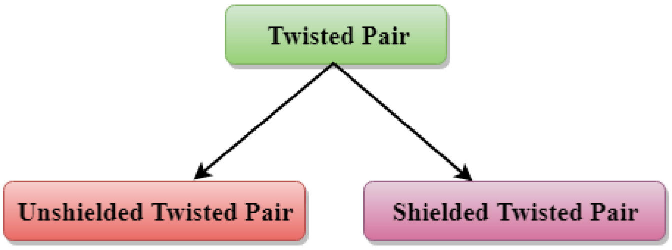
Types of Guided media:

1. **Twisted pair:**

- Twisted pair is a physical media made up of a pair of cables twisted with each other.
- A twisted pair cable is cheap as compared to other transmission media. Installation of the twisted pair cable is easy, and it is a lightweight cable.
- The frequency range for twisted pair cable is from 0 to 3.5KHz.
- A twisted pair consists of two insulated copper wires arranged in a regular spiral pattern.
- The degree of reduction in noise interference is determined by the number of turns per foot. Increasing the number of turns per foot decreases noise interference.



Types of Twisted pair:



A. Unshielded Twisted Pair:

An unshielded twisted pair is widely used in telecommunication. Following are the categories of the unshielded twisted pair cable:

- **Category 1:** Category 1 is used for telephone lines that have low-speed data.
- **Category 2:** It can support upto 4Mbps.
- **Category 3:** It can support upto 16Mbps.
- **Category 4:** It can support upto 20Mbps. Therefore, it can be used for long-distance communication.
- **Category 5:** It can support upto 200Mbps.

Advantages of Unshielded Twisted Pair:

- It is cheap.
- Installation of the unshielded twisted pair is easy.
- It can be used for high-speed LAN.

Disadvantage:

- This cable can only be used for shorter distances because of attenuation.

B. Shielded Twisted Pair

A shielded twisted pair is a cable that contains the mesh surrounding the wire that allows the higher transmission rate.

Characteristics of Shielded Twisted Pair:

- The cost of the shielded twisted pair cable is not very high and not very low.
- An installation of STP is easy.
- It has higher capacity as compared to unshielded twisted pair cable.
- It has a higher attenuation.
- It is shielded that provides the higher data transmission rate.

Disadvantages

- It is more expensive as compared to UTP and coaxial cable.
- It has a higher attenuation rate.

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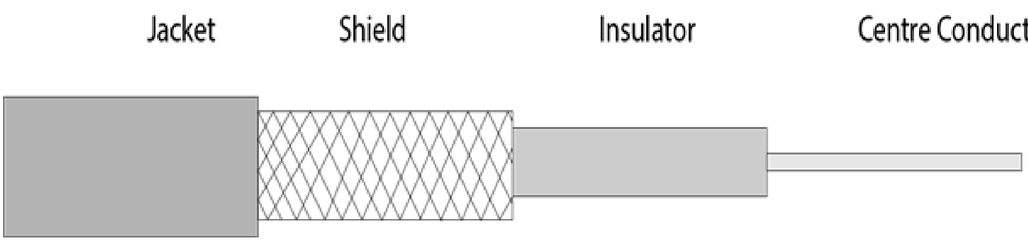
2. Coaxial Cable

- Coaxial cable is very commonly used transmission media, for example, TV wire is usually a coaxial cable.
- The name of the cable is coaxial as it contains two conductors parallel to each other.
- It has a higher frequency as compared to Twisted pair cable.
- The inner conductor of the coaxial cable is made up of copper, and the outer conductor is made up of copper mesh. The middle core is made up of non-conductive cover that separates the inner conductor from the outer conductor.
- The middle core is responsible for the data transferring whereas the copper mesh prevents from the **EMI** (Electromagnetic interference).



Coaxial cable is of two types:

1. **Baseband transmission:** It is defined as the process of transmitting a single signal at high speed.
2. **Broadband transmission:** It is defined as the process of transmitting multiple signals simultaneously.

Advantages of Coaxial cable:

- The data can be transmitted at high speed.
- It has better shielding as compared to twisted pair cable.
- It provides higher bandwidth.

Disadvantages of Coaxial cable:

- It is more expensive as compared to twisted pair cable.
- If any fault occurs in the cable causes the failure in the entire network.

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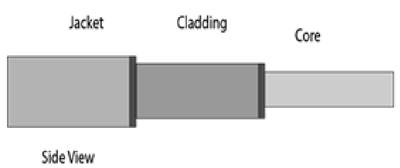
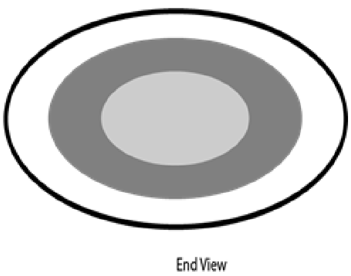
3. Fibre Optic

- Fibre optic cable is a cable that uses electrical signals for communication.
- Fibre optic is a cable that holds the optical fibres coated in plastic that are used to send the data by pulses of light.
- The plastic coating protects the optical fibres from heat, cold, electromagnetic interference from other types of wiring.
- Fibre optics provide faster data transmission than copper wires.

Diagrammatic representation of fibre optic cable:





Basic elements of Fibre optic cable:

- **Core:** The optical fibre consists of a narrow strand of glass or plastic known as a core. A core is a light transmission area of the fibre. The more the area of the core, the more light will be transmitted into the fibre.
- **Cladding:** The concentric layer of glass is known as cladding. The main functionality of the cladding is to provide the lower refractive index at the core interface as to cause the reflection within the core so that the light waves are transmitted through the fibre.
- **Jacket:** The protective coating consisting of plastic is known as a jacket. The main purpose of a jacket is to preserve the fibre strength, absorb shock and extra fibre protection.

Following are the advantages of fibre optic cable over copper:

- **Greater Bandwidth:** The fibre optic cable provides more bandwidth as compared copper. Therefore, the fibre optic carries more data as compared to copper cable.
- **Faster speed:** Fibre optic cable carries the data in the form of light. This allows the fibre optic cable to carry the signals at a higher speed.
- **Longer distances:** The fibre optic cable carries the data at a longer distance as compared to copper cable.
- **Better reliability:** The fibre optic cable is more reliable than the copper cable as it is immune to any temperature changes while it can cause obstruct in the connectivity of copper cable.
- **Thinner and Sturdier:** Fibre optic cable is thinner and lighter in weight so it can withstand more pull pressure than copper cable.

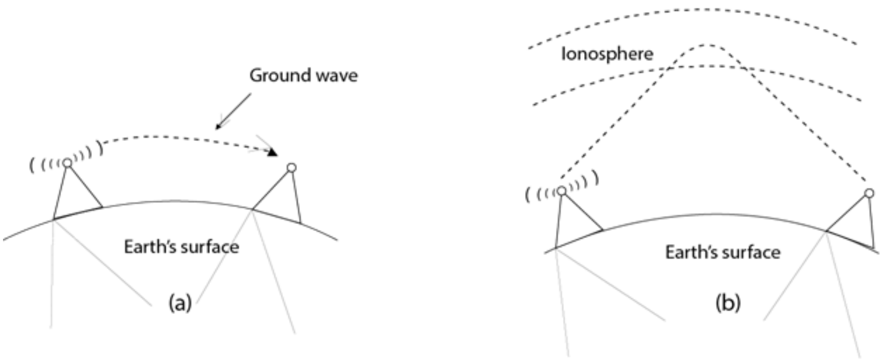
Unguided Transmission

- An unguided transmission transmits the electromagnetic waves without using any physical medium. Therefore, it is also known as **wireless transmission**.
- In unguided media, air is the media through which the electromagnetic energy can flow easily.

Unguided transmission is broadly classified into three categories:

1) **Radio waves**

- Radio waves are the electromagnetic waves that are transmitted in all the directions of free space.
- Radio waves are omnidirectional, i.e., the signals are propagated in all the directions.
- The range in frequencies of radio waves is from 3Khz to 1 khz.
- In the case of radio waves, the sending and receiving antenna are not aligned, i.e., the wave sent by the sending antenna can be received by any receiving antenna.
- An example of the radio wave is **FM radio**.



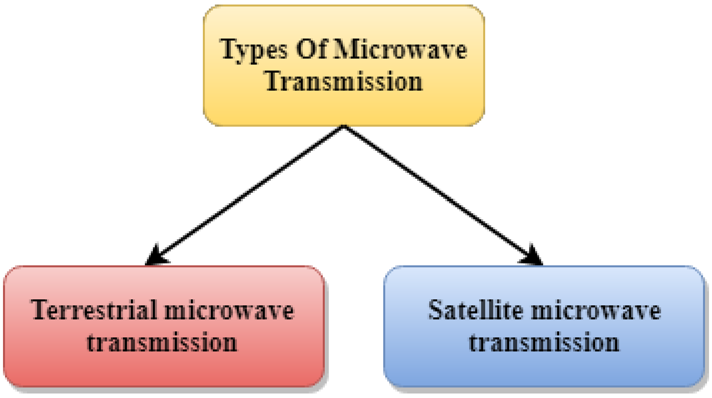
Applications of Radio waves:

- A Radio wave is useful for multicasting when there is one sender and many receivers.
- An FM radio, television, cordless phones are examples of a radio wave.

Advantages of Radio transmission:

- Radio transmission is mainly used for wide area networks and mobile cellular phones.
- Radio waves cover a large area, and they can penetrate the walls.
- Radio transmission provides a higher transmission rate.

2) Microwaves



Terrestrial Microwave Transmission

- Terrestrial Microwave transmission is a technology that transmits the focused beam of a radio signal from one ground-based microwave transmission antenna to another.
- Microwaves are the electromagnetic waves having the frequency in the range from 1GHz to 1000 GHz.
- Microwaves are unidirectional as the sending and receiving antenna is to be aligned, i.e., the waves sent by the sending antenna are narrowly focussed.
- In this case, antennas are mounted on the towers to send a beam to another antenna which is km away.
- It works on the line of sight transmission, i.e., the antennas mounted on the towers are the direct sight of each other.

Characteristics of Microwave:

- **Frequency range:** The frequency range of terrestrial microwave is from 4-6 GHz to 21-23 GHz.
- **Bandwidth:** It supports the bandwidth from 1 to 10 Mbps.
- **Short distance:** It is inexpensive for short distance.
- **Long distance:** It is expensive as it requires a higher tower for a longer distance.
- **Attenuation:** Attenuation means loss of signal. It is affected by environmental conditions and antenna size.

Advantages of Microwave:

- Microwave transmission is cheaper than using cables.
- It is free from land acquisition as it does not require any land for the installation of cables.
- Microwave transmission provides an easy communication in terrains as the installation of cable in terrain is quite a difficult task.
- Communication over oceans can be achieved by using microwave transmission.

Disadvantages of Microwave transmission:

- **Eavesdropping:** An eavesdropping creates insecure communication. Any malicious user can catch the signal in the air by using its own antenna.
- **Out of phase signal:** A signal can be moved out of phase by using microwave transmission.
- **Susceptible to weather condition:** A microwave transmission is susceptible to weather condition. This means that any environmental change such as rain, wind can distort the signal.
- **Bandwidth limited:** Allocation of bandwidth is limited in the case of microwave transmission.

3) Satellite Microwave Communication

- A satellite is a physical object that revolves around the earth at a known height.
- Satellite communication is more reliable nowadays as it offers more flexibility than cable and fibre optic systems.
- We can communicate with any point on the globe by using satellite communication.

How Does Satellite work?

The satellite accepts the signal that is transmitted from the earth station, and it amplifies the signal. The amplified signal is retransmitted to another earth station.

Advantages of Satellite Microwave Communication:

- The coverage area of a satellite microwave is more than the terrestrial microwave.
- The transmission cost of the satellite is independent of the distance from the centre of the coverage area.
- Satellite communication is used in mobile and wireless communication applications.
- It is easy to install.
- It is used in a wide variety of applications such as weather forecasting, radio/TV signal broadcasting, mobile communication, etc.

Disadvantages of Satellite Microwave Communication:

- Satellite designing and development requires more time and higher cost.
- The Satellite needs to be monitored and controlled on regular periods so that it remains in orbit.
- The life of the satellite is about 12-15 years. Due to this reason, another launch of the satellite has to be planned before it becomes non-functional.

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4) Infrared

- An infrared transmission is a wireless technology used for communication over short ranges.
- The frequency of the infrared in the range from 300 GHz to 400 THz.
- It is used for short-range communication such as data transfer between two cell phones, TV remote operation, data transfer between a computer and cell phone resides in the same closed area.

Characteristics of Infrared:

- It supports high bandwidth, and hence the data rate will be very high.
- Infrared waves cannot penetrate the walls. Therefore, the infrared communication in one room cannot be interrupted by the nearby rooms.

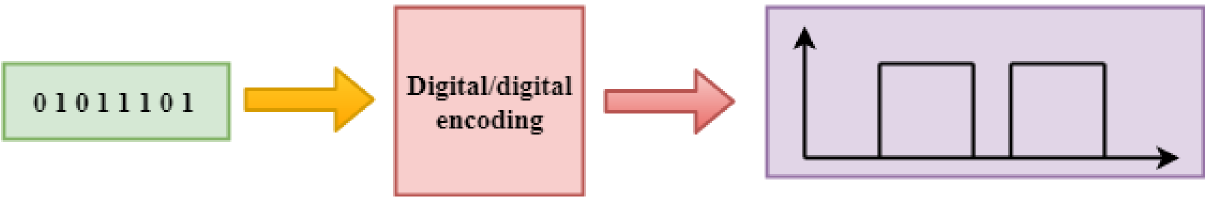
- An infrared communication provides better security with minimum interference.
- Infrared communication is unreliable outside the building because the sun rays will interfere with the infrared waves.

Digital Transmission

Data can be represented either in analog or digital form. The computers used the digital form to store the information. Therefore, the data needs to be converted in digital form so that it can be used by a computer.

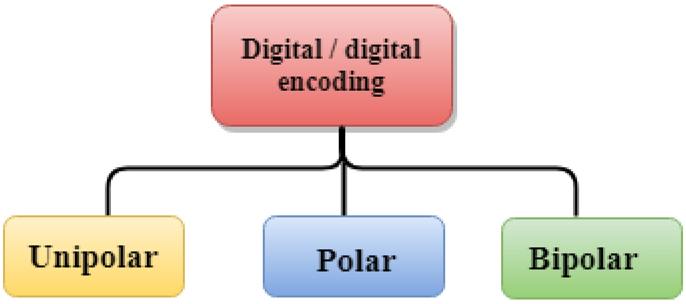
- DIGITAL-TO-DIGITAL CONVERSION

Digital-to-digital encoding is the representation of digital information by a digital signal. When binary 1s and 0s generated by the computer are translated into a sequence of voltage pulses that can be propagated over a wire, this process is known as digital-to-digital encoding.



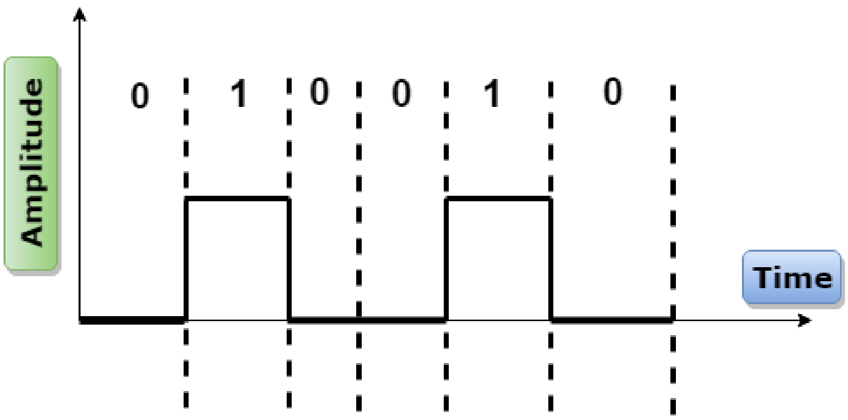
Digital-to-digital encoding is divided into three categories:

- Unipolar Encoding
- Polar Encoding
- Bipolar Encoding



Unipolar

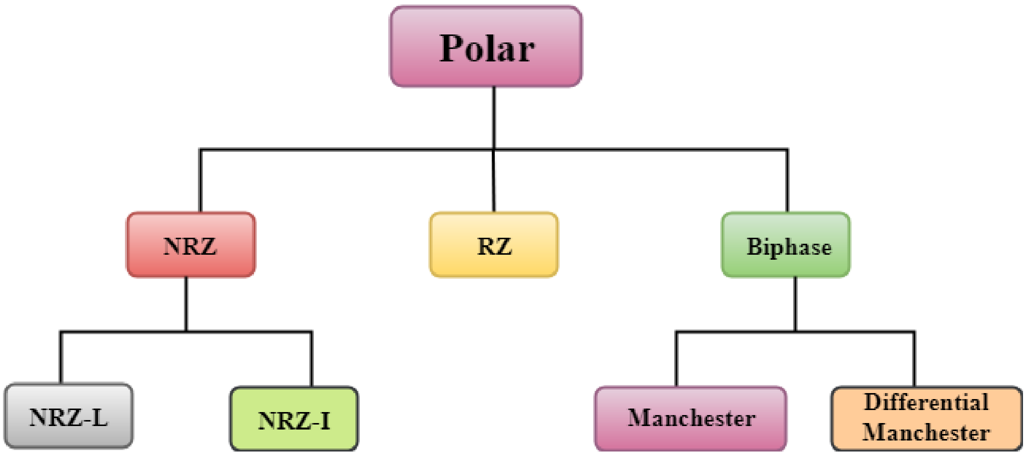
- Digital transmission system sends the voltage pulses over the medium link such as wire or cable.
- In most types of encoding, one voltage level represents 0, and another voltage level represents 1.
- The polarity of each pulse determines whether it is positive or negative.
- This type of encoding is known as Unipolar encoding as it uses only one polarity.
- In Unipolar encoding, the polarity is assigned to the 1 binary state.
- In this, 1s are represented as a positive value and 0s are represented as a zero value.
- In Unipolar Encoding, '1' is considered as a high voltage and '0' is considered as a zero voltage.
- Unipolar encoding is simpler and inexpensive to implement.



- Unipolar encoding has two problems that make this scheme less desirable: DC Component
- Synchronization

Polar

- Polar encoding is an encoding scheme that uses two voltage levels: one is positive, and another is negative.
- By using two voltage levels, an average voltage level is reduced, and the DC component problem of unipolar encoding scheme is alleviated.

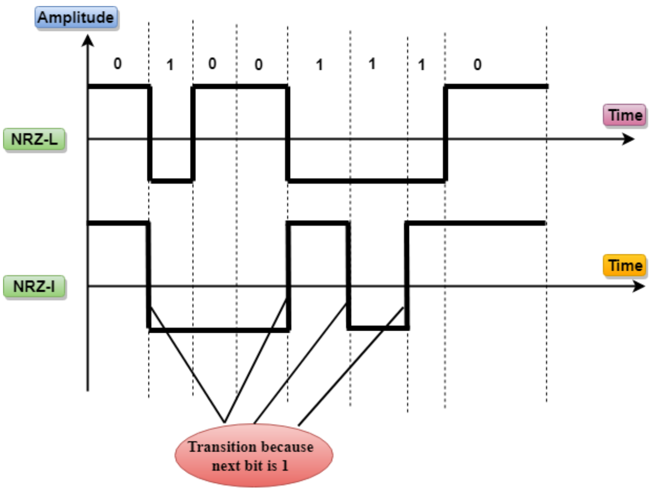


- **NRZ**
  - NRZ stands for Non-return zero.
  - In NRZ encoding, the level of the signal can be represented either positive or negative.

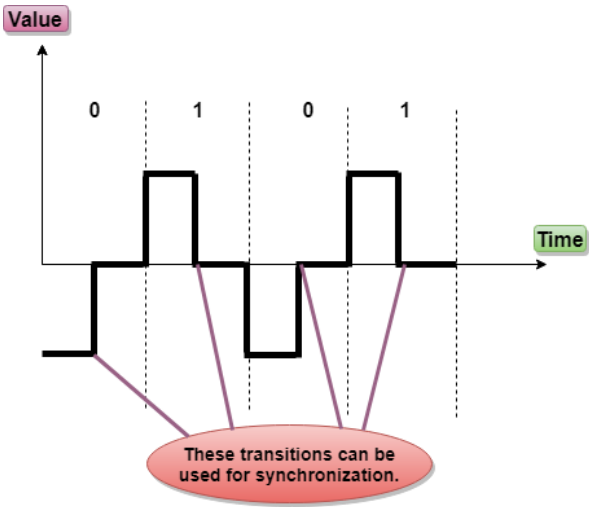
The two most common methods used in NRZ are:

1.NRZ-L: In NRZ-L encoding, the level of the signal depends on the type of the bit that it represents. If a bit is 0 or 1, then their voltages will be positive and negative respectively. Therefore, we can say that the level of the signal is dependent on the state of the bit.

2.NRZ-I: NRZ-I is an inversion of the voltage level that represents 1 bit. In the NRZ-I encoding scheme, a transition occurs between the positive and negative voltage that represents 1 bit. In this scheme, 0 bit represents no change and 1 bit represents a change in voltage level.



- **RZ**
  - RZ stands for Return to zero.
  - There must be a signal change for each bit to achieve synchronization. However, to change with every bit, we need to have three values: positive, negative and zero.
  - RZ is an encoding scheme that provides three values, positive voltage represents 1, the negative voltage represents 0, and zero voltage represents none.
  - In the RZ scheme, halfway through each interval, the signal returns to zero.
  - In RZ scheme, 1 bit is represented by positive-to-zero and 0 bit is represented by negative-to-zero.



Disadvantage of RZ:

It performs two signal changes to encode one bit that acquires more bandwidth.

- **Biphase**
  - Biphase is an encoding scheme in which signal changes at the middle of the bit interval but does not return to zero.

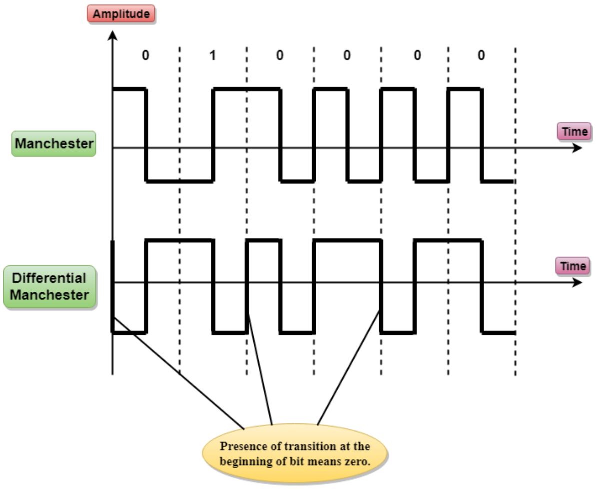
Biphase encoding is implemented in two different ways:

1.Manchester

- It changes the signal at the middle of the bit interval but does not return to zero for synchronization.
- In Manchester encoding, a negative-to-positive transition represents binary 1, and positive-to-negative transition represents 0.
- Manchester has the same level of synchronization as RZ scheme except that it has two levels of amplitude.

2.Differential Manchester

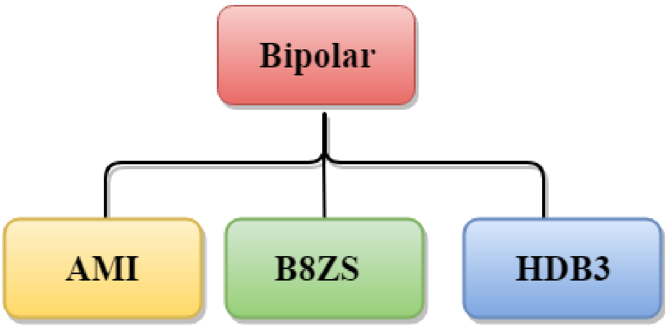
- It changes the signal at the middle of the bit interval for synchronization, but the presence or absence of the transition at the beginning of the interval determines the bit. A transition means binary 0 and no transition means binary 1.
- In Manchester Encoding scheme, two signal changes represent 0 and one signal change represent 1.



➢ Bipolar

- Bipolar encoding scheme represents three voltage levels: positive, negative, and zero.
- In Bipolar encoding scheme, zero level represents binary 0, and binary 1 is represented by alternating positive and negative voltages.
- If the first 1 bit is represented by positive amplitude, then the second 1 bit is represented by negative voltage, third 1 bit is represented by the positive amplitude and so on. This alternation can also occur even when the 1bits are not consecutive.

Bipolar can be classified as:



1.AMI

- AMI stands for ***alternate mark inversion*** where mark work comes from telegraphy which means 1. So, it can be redefined as **alternate 1 inversion**.
- In Bipolar AMI encoding scheme, 0 bit is represented by zero level and 1 bit is represented by alternating positive and negative voltages.

Advantage:

- DC component is zero.
- Sequence of 1s bits are synchronized.

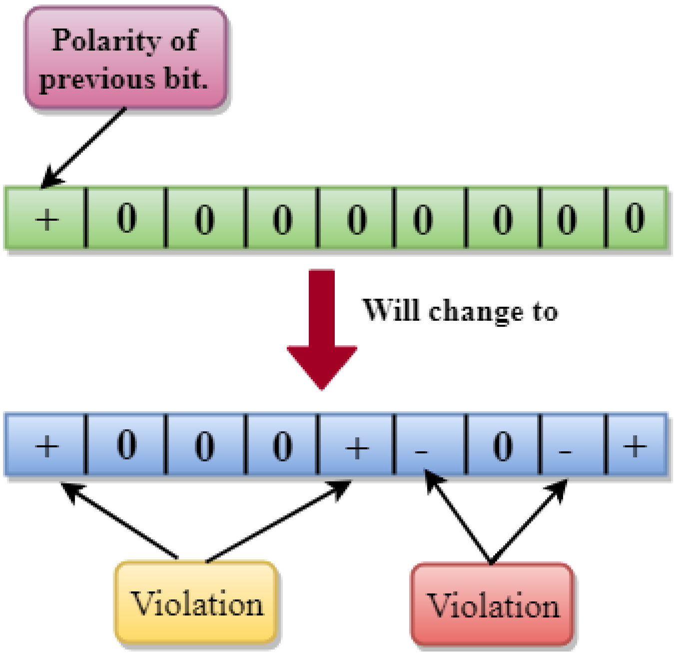
Disadvantage:

- This encoding scheme does not ensure the synchronization of a long string of 0s bits.

2.B8ZS

- B8ZS stands for **Bipolar 8-Zero Substitution**.
- This technique is adopted in North America to provide synchronization of a long sequence of 0s bits.
- In most of the cases, the functionality of B8ZS is similar to the bipolar AMI, but the only difference is that it provides the synchronization when a long sequence of 0s bits occur.
- B8ZS ensures synchronization of a long string of 0s by providing force artificial signal changes called violations, within 0 string pattern.
- When eight 0 occurs, then B8ZS implements some changes in 0s string pattern based on the polarity of the previous 1 bit.

- If the polarity of the previous 1 bit is positive, the eight 0s will be encoded as zero, zero, zero, positive, negative, zero, negative, positive.

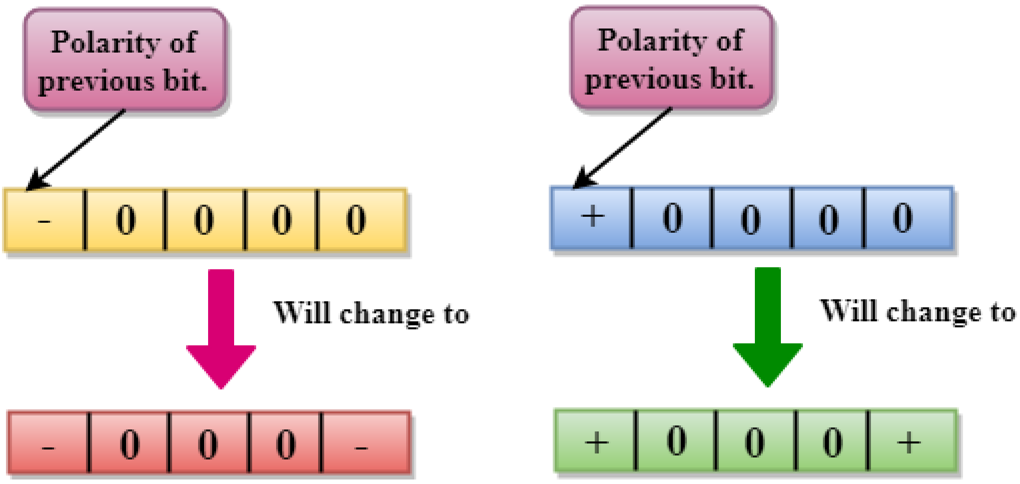


- If the polarity of previous 1 bit is negative, then the eight 0s will be encoded as zero, zero, zero, negative, positive, zero, positive, negative.

3.HDB3

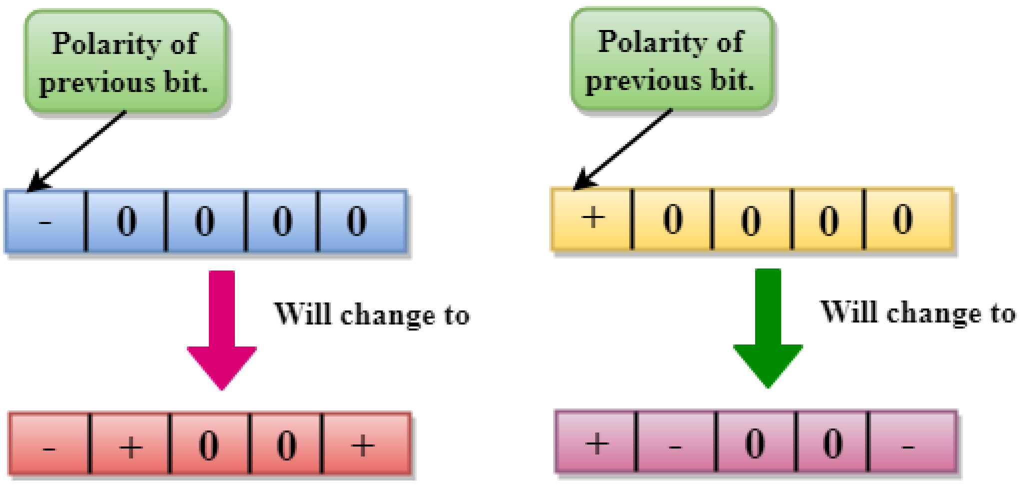
- HDB3 stands for **High-Density Bipolar 3**.
- HDB3 technique was first adopted in Europe and Japan.
- HDB3 technique is designed to provide the synchronization of a long sequence of 0s bits.
- In the HDB3 technique, the pattern of violation is based on the polarity of the previous bit.
- When four 0s occur, HDB3 looks at the number of 1s bits occurred since the last substitution.
- If the number of 1s bits is odd, then the violation is made on the fourth consecutive of 0. If the polarity of the previous bit is positive, then the violation is positive. If the polarity of the previous bit is negative, then the violation is negative.

If the number of 1s bits since the last substitution is odd.



If the number of 1s bits is even, then the violation is made on the place of the first and fourth consecutive 0s. If the polarity of the previous bit is positive, then violations are negative, and if the polarity of the previous bit is negative, then violations are positive.

If the number of 1s bits since the last substitution is even.



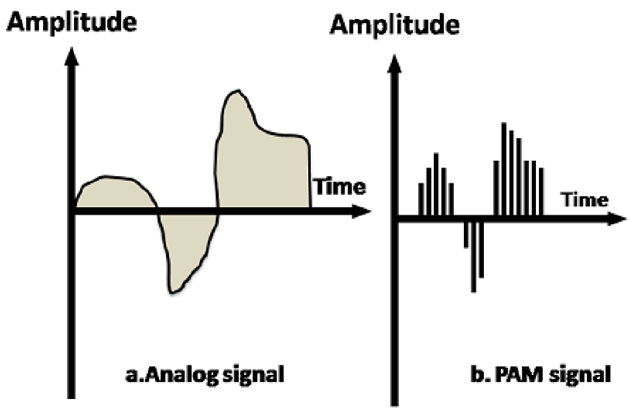
- **ANALOG-TO-DIGITAL CONVERSION**
  - When an analog signal is digitalized, this is called an analog-to-digital conversion.
  - Suppose human sends a voice in the form of an analog signal, we need to digitalize the analog signal which is less prone to noise. It requires a reduction in the number of values in an analog message so that they can be represented in the digital stream.

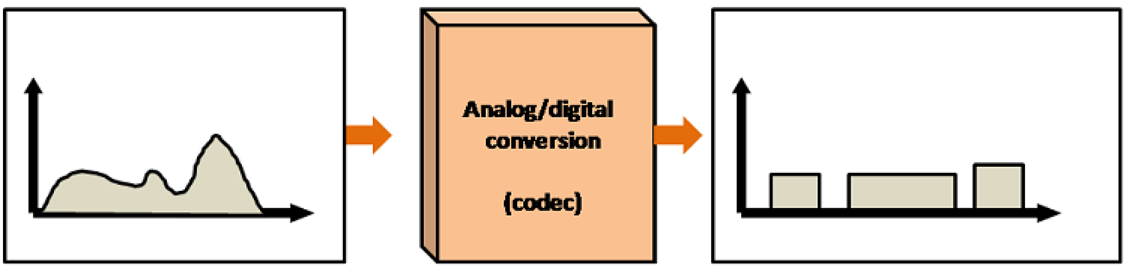
- In analog-to-digital conversion, the information contained in a continuous wave form is converted in digital pulses.

Techniques for Analog-To-Digital Conversion

PAM

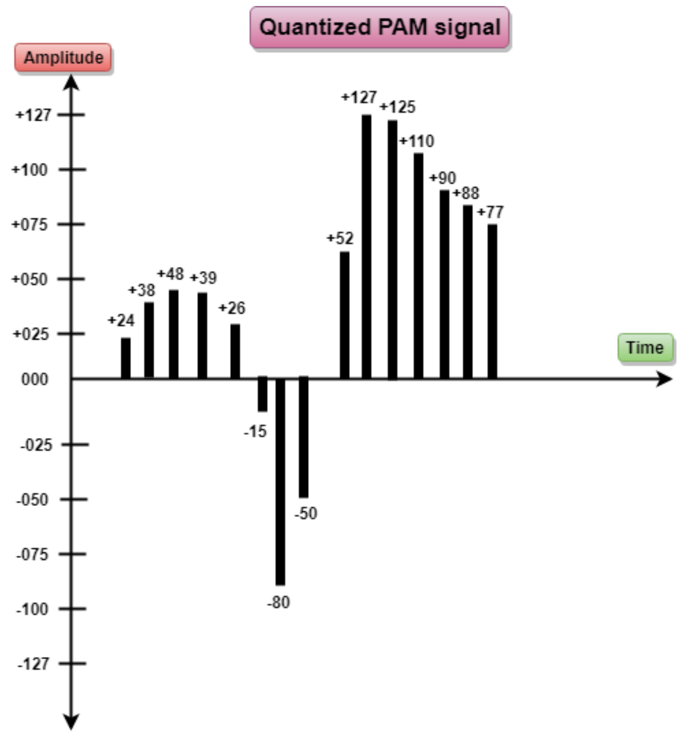
- PAM stands for **pulse amplitude modulation**.
- PAM is a technique used in analog-to-digital conversion.
- PAM technique takes an analog signal, samples it, and generates a series of digital pulses based on the result of sampling where sampling means measuring the amplitude of a signal at equal intervals.
- PAM technique is not useful in data communication as it translates the original wave form into pulses, but these pulses are not digital. To make them digital, PAM technique is modified to PCM technique.





PCM

- PCM stands for **Pulse Code Modulation**.
- PCM technique is used to modify the pulses created by PAM to form a digital signal. To achieve this, PCM quantizes PAM pulses. Quantization is a process of assigning integral values in a specific range to sampled instances.
- PCM is made of four separate processes: PAM, quantization, binary encoding, and digital-to-digital encoding.

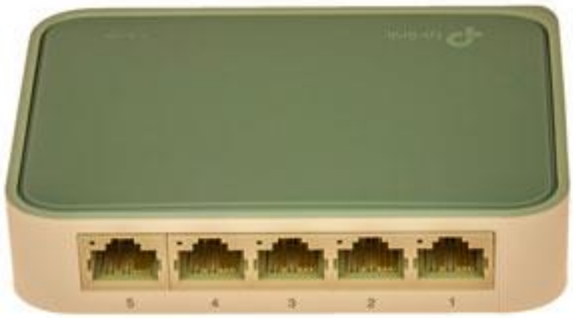


PCM

What is an Ethernet Switch & How It works?

- Ethernet switches are the key element in most data networks. Although networks switches could use any one of a variety of communications standards, Ethernet is by far the most widely used.
- An Ethernet switch can be small, or very large - major networks are known to have failed when a network switch has gone down. This shows the key nature of network switches and in this case, Ethernet switches within networks.

- Switches are also used for industrial control, where robustness and carrier grade performance are required - naturally these industrial switches have a price to reflect this and are not normally used where standard switches can suffice.



What is an Ethernet switch?

- Network switches and in this case, Ethernet switches are key building blocks for any network. These switches connect multiple devices together when they are on the same network.
- The nodes on the network will include devices such as: computers, servers; wireless access points, printers, and so forth.
- The Ethernet switch enables the connected devices on the network to exchange data and talk to one another other.
- Network switches are semi-intelligent devices that enable the data to be routed in the required direction so that when a packet of data is sent from the source to the destination it is routed correctly.
- An Ethernet network switch is able to work with the MAC addresses of the devices connected to it. Using this information, it is able to identify the computers or other units on each of its ports. In this way it is able to send the data packets to the relevant ports and hence to the right devices without flooding the network with unnecessary data.
- Additionally, an Ethernet switch is able to allocate the full bandwidth to each of its ports. This means that regardless of the number of devices operating, users will always have access to the maximum amount of bandwidth.

Managed & unmanaged Ethernet switches

When looking at the descriptions of Ethernet switches, it will be seen that some of them are referred to as unmanaged switches, whereas others are referred to as being managed switches. When selecting an Ethernet switch it is important to select the required type.

- ***Unmanaged switches:*** An unmanaged network switch is the most common form of Ethernet switch. This type of Ethernet switch is design to be simply plugged into the network and then operate without any manual configuration. Unmanaged switches are typically for basic connectivity and they are often used in home or small office networks or wherever a few more Ethernet ports are needed, at a desk, in a lab, in a conference room, etc.
- ***Managed switches:*** Managed switches give greater security with more features and flexibility because they can be configured to custom-fit the network. With this greater control, it is possible to better protect the network and improve the quality of service for those who access the network. The traffic can be prioritised so that the available bandwidth, etc is allocated to a given application, etc in the best way.

There is also a type of switch referred to as an **industrial switch**. They are typically used for industrial control, and similar applications. They are called industrial switches because they are used within industrial environments requiring a high level of robustness and tolerance of wide temperature ranges etc, in addition to this industrial switch feature carrier grade Ethernet performance because industrial and production environments need to have very high reliability in view of the costs of any disruption

How Ethernet switches work - the basics

Ethernet switches are used to link various Ethernet devices together, relaying the data from one device on the data network such as a local area network to another. These devices may even be other Ethernet switches, or devices such as computers, servers, printers, etc.

When moving data around an Ethernet based network, data frames called Ethernet frames are used.

The switch uses these frames and relays them between the devices that are connected to them.

By moving Ethernet frames between the switch ports in this way, a switch is able to link the traffic carried by the individual network connections into a larger Ethernet network such as a local area network, etc.

Ethernet switch for use within local area networks

Ethernet switches work by "bridging" Ethernet frames between different segments of a local area network.

The switch performs its function by copying the frames from one switch port to another according to the Media Access Control, MAC addresses int he frames.

It is worth noting that the concept of Ethernet bridging was defined in IEEE 802.1D, the IEEE standard for Metropolitan and Local Area Networks: Media Access Control Bridges.

The fact that there is a universally adopted standard for this operation means that Ethernet switches bought from different manufacturers can all interface together and work in any Ethernet based local area network, or even larger networks like Metropolitan Area Networks or Wide Area Networks.

The Ethernet frames are used to transport the data over the Ethernet cables, and lines between the nodes on the network. Essentially frames have a defined format and they are sequences of information, containing the data to be transported, but with additional data or information around it to manage the sending and receiving of the data successfully.

Basic overall Ethernet frame format

The format of the Ethernet frame includes a number of different elements as seen in the diagram:

- ***Header:*** This consists of a preamble to enable the receiver to synchronise, a destination address, a source address and a Length / type indication.
- ***Payload:*** This is the actual data to be carried around the local area network by the Ethernet frame.

- ***Trailer:*** This consists of a Frame Check Sequence for error checking.

The basic format for these Ethernet frames has remained the same over the many versions of the Ethernet standard. With Ethernet data travelling over many local area networks and WAN networks, consisting of equipment utilising many different versions of the standard: 100Mbps Fast Ethernet, Gigabit Ethernet 1GE, 10 GE, etc, the frames provide an essential and constant format for this data, allowing backward compatibility.

Ethernet switch operation

One of the key aspects of Ethernet switches is that their operation should appear transparent to the devices with which they connect. Accordingly, this type of operation is called "transparent bridging."

For transparent bridging, the Ethernet frames remain the same as they pass through the switch. As a result, the switch will automatically start to work without needing any configuration on the switch or the various nodes that are connected to the Ethernet network, and this makes the operation of the switch transparent to them all.

One of the key elements of an Ethernet switch is the address learning which it does. The device does not forward all frames onto all its ports - if it did this then the network would soon become overloaded. Instead the switch forwards the relevant frames to the relevant ports.

The switch achieves this by using the traffic forwarding rules defined in he IEEE 802.1D bridging standard.

Under this standard, the traffic forwarding is based around the switch learning where to send the data. The switch makes the decisions about where to forward the data based upon the 48-bit Media Access Control address contained within the Ethernet frame.

In order to route the data in the right direction, the switch learns which devices (called stations in the standard) are on which segments of the network. It does this by looking at the source addresses in the Ethernet frames it receives.

All Ethernet interfaces have a unique MAC address which is assigned during manufacture. Normally an Ethernet device will only accept frames that are directed to it, but in the case of an Ethernet switch, this is not the case because it handles frames that are intended for other devices - it runs in what is often termed "promiscuous mode."

As each frame is received on a given port, the software within the Ethernet switch looks at the source address contained within the frame and adds this to its table of addresses that it maintains within the memory of the switch. In this way it remembers which devices are attached to it, so that it can correctly route frames to the required destination.

The list is effectively a database and this enables the switch to make a packet forwarding decision for each frame that is received in a process called adaptive filtering.

This means that only the Ethernet frames intended for a particular device are sent along that path, and in this way the level of data being carried around the local area network is minimised.

Ethernet switch traffic management aspects

There are several aspects to the way in which an Ethernet switch handles the traffic. There are several features and capabilities that are key.

- ***Traffic filtering & forwarding:*** With the addresses of the different stations or devices connected to the different ports of the Ethernet switch, it is able to forward the various frames it receives to the required destinations.
  As devices connected to the switch may not be ready to receive frames, or the switch itself may be busy, the different ports have the ability to hold frames until the system is ready for them to be sent.

  Also, and frames received are forwarded on without any change, making the switch totally transparent to the stations on the local area network, or WAN network or whatever form of network it is.

- ***Frame Flooding:*** As will sometimes happen, the Ethernet switch will receive a frame that is destined for a station or device that is new. When this happens, the switch sends out a frame to all the ports, other than the one on which it received the frame with the unknown destination to discover the station which has the required address.

<u>What is MAC Address?</u>

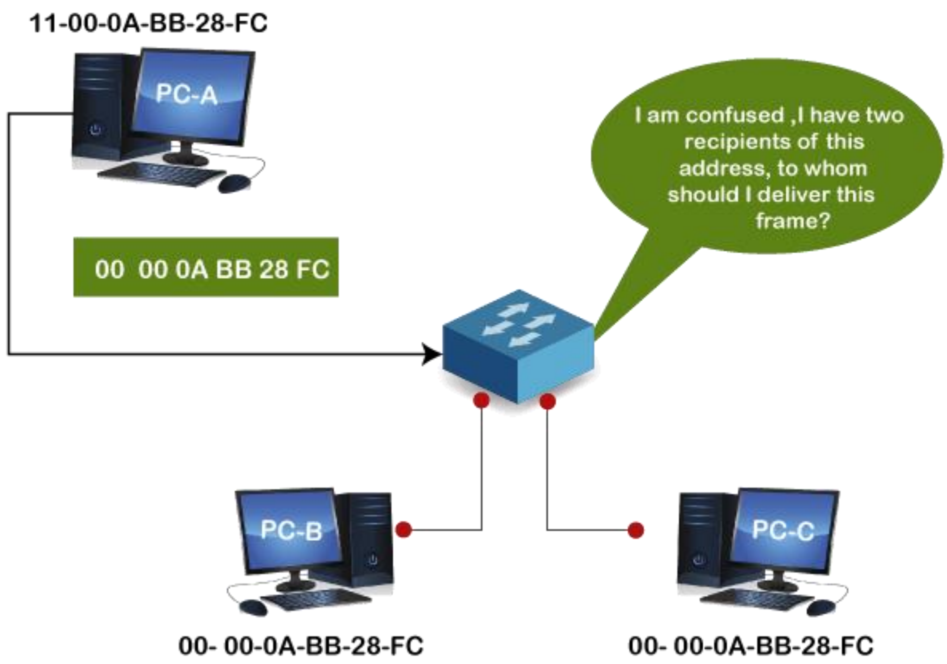
- MAC address is the physical address, which uniquely identifies each device on a given network. To make communication between two networked devices, we need two addresses: **IP address and MAC address.** It is assigned to the NIC (Network Interface card) of each device that can be connected to the internet.
- It stands for **Media Access Control**, and also known as **Physical address, hardware address, or BIA (Burned in Address).**
- It is globally unique; it means two devices cannot have the same MAC address. It is represented in a hexadecimal format on each device, such as **00:0a:95:9d:67:16.**
- It is 12-digit, and 48 bits long, out of which the first *24 bits are used for **OUI** (Organization Unique Identifier),* and *24 bits are for NIC/vendor-specific.*
- It works on the data link layer of the OSI model.
- It is provided by the device's vendor at the time of manufacturing and embedded in its NIC, which is ideally cannot be changed.
- The **ARP protocol** is used to associate a logical address with a physical or MAC address.

Why should the MAC address be unique in the LAN network?

If a LAN network has two or more devices with the same MAC address, that network will not work.

Suppose three devices A, B, and C are connected to a network through a switch. The MAC addresses of these devices are 11000ABB28FC, 00000ABB28FC, and 00000ABB28FC, respectively. The NIC of devices B and C have the same MAC address. If device A sends a data frame to the address 00000ABB28FC, the switch will fail to deliver this frame to the destination, as it has two recipients of this data frame.

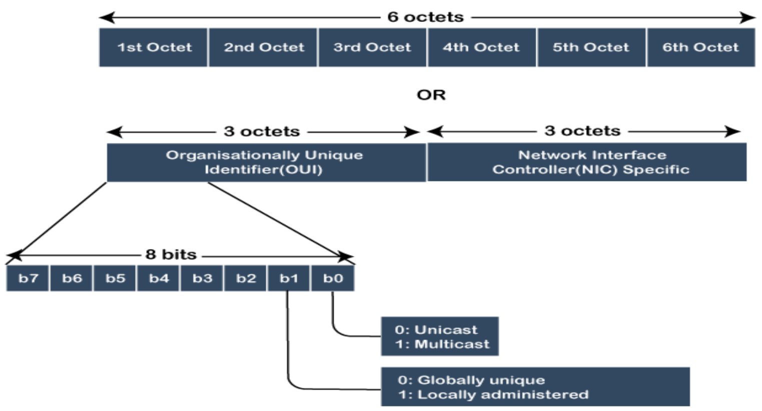
We can understand this example with the below image:



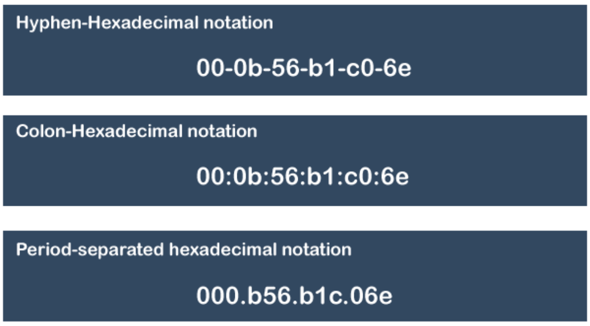
Format of MAC address

- It is 12 digits or 6-byte hexadecimal number, which is represented in colon-hexadecimal notation format. It is divided into six octets, and each octet contains 8 bits.
- The first three octets are used as the **OUI or Organisationally Unique Identifier**. These MAC prefixes are assigned to each organization or vendor by the IEEE Registration Authority Committee.

Some example of OUI of known vendors are:
CC: 46: D6 - Cisco
3C: 5A: B4 - Google, Inc.
3C: D9: 2B - Hewlett Packard
00: 9A: CD - HUAWEI TECHNOLOGIES CO., LTD



- The last three octets are NIC specific and used by the manufacturer to each NIC card. Vendors or manufacturers can use any sequence of digits to the NIC specific digits, but the prefix should be the same as provided by the IEEE.
- The MAC address can be represented in below three formats:



Types of MAC address

There are three types of MAC addresses, which are:

1. **Unicast MAC Address**
2. **Multicast MAC address**
3. **Broadcast MAC address**

Unicast MAC address:

The Unicast MAC address represents the specific NIC on the network. A Unicast MAC address frame is only sent out to the interface which is assigned to a specific NIC and hence transmitted to the single destination device. If the LSB (least significant bit) of the first octet of an address is set to zero, the frame is meant to reach only one destination NIC.

Multicast MAC Address:

Multicast addresses enables the source device to transmit a data frame to multiple devices or NICs. In Layer-2 (Ethernet) Multicast address, LSB (least significant bit) or first 3 bytes of the first octet of an address is set to one and reserved for the multicast addresses. The rest 24 bits are used by the device that wants to send the data in a group. The multicast address always starts with the prefix 01-00-5E.

Broadcast MAC address

It represents all devices within a Network. In broadcast MAC address, Ethernet frames with ones in all bits of the destination address (FF-FF-FF-FF-FF-FF) are known as a b**roadcast address**. All these bits are the reserved addresses for the broadcast. Frames that are destined with MAC address FF-FF-FF-FF-FF-FF will reach every computer belong to that LAN segment. Hence if a source device wants to send the data to all the devices within a network, that can use the broadcast address as the destination MAC address.

Address Resolution Protocol (ARP)

Address Resolution Protocol (ARP) is a communication protocol used to find the MAC (Media Access Control) address of a device from its IP address. This protocol is used when a device wants to communicate with another device on a Local Area Network or Ethernet.

Types of ARP

- Proxy ARP
- Gratuitous ARP
- Reverse ARP (RARP)
- Inverse ARP

1.Proxy ARP - Proxy ARP is a method through which a Layer 3 devices may respond to ARP requests for a target that is in a different network from the sender. The proxy ARP configured router responds to the ARP and map the MAC address of the router with the target ip address and fool the sender that it is reached at its destination. At the backend, the proxy router sends its packets to the appropriate destination because the packets contain the necessary information.

Example - If Host A wants to transmit data to Host B, which is on the different network, then Host A sends an ARP request message to receive a MAC address for Host B. The router responds to Host A with its own MAC address pretend itself as a destination. When the data is transmitted to the destination by Host A, it will send to the gateway so that it sends to Host B. This is known as proxy ARP.

2.Gratuitous ARP - Gratuitous ARP is an ARP Request of the host that helps to identify the duplicate IP address. It is a broadcast request for the IP address of the router. If an ARP request is sent by a switch or router to get its IP address and no ARP responses are received, so all other nodes cannot use the IP address allocated to that switch or router. Yet if a router or switch sends an ARP request for its IP address and receives an ARP response, another node uses the IP address allocated to the switch or router.

There are some primary use cases of gratuitous ARP that are given below:

- The gratuitous ARP is used to update the ARP table of other devices.
- It also checks whether the host is using the original IP address or a duplicate one.

3.Reverse ARP (RARP) - It is a networking protocol used by the client system in a local area network (LAN) to request its IPv4 address from the ARP gateway router table. A table is created by the network administrator in the gateway-router that is used to find out the MAC address to the corresponding IP address. When a new system is set up or any machine that has no memory to store the IP address, then the user has to find the IP address of the device. The device sends a RARP broadcast packet, including its own MAC address in the address field of both the sender and the receiver hardware. A host installed inside of the local network called the RARP-server is prepared to respond to such type of broadcast packet. The RARP server is then trying to locate a mapping table entry in the IP to MAC address. If any entry matches the item in the table, then the RARP server sends the response packet along with the IP address to the requesting computer.

4.Inverse ARP (InARP) - Inverse ARP is inverse of the ARP, and it is used to find the IP addresses of the nodes from the data link layer addresses. These are mainly used for the frame relays, and ATM networks, where Layer 2 virtual circuit addressing are often acquired from Layer 2 signaling. When using these virtual circuits, the relevant Layer 3 addresses are available. ARP conversions Layer 3 addresses to Layer 2 addresses. However, its opposite address can be defined by InARP. The InARP has a similar packet format as ARP, but operational codes are different.

Forwarding Information Base (FIB)

- A **forwarding information base** (**FIB**), also known as a **forwarding table** or **MAC table**.
- Most commonly used in network bridging, routing, and similar functions to find the proper output network interface controller to which the input interface should forward a packet.
- It is a dynamic table that maps MAC addresses to ports.
- It is the essential mechanism that separates network switches from Ethernet hubs.
- Content-addressable memory (CAM) is typically used to efficiently implement the FIB, thus it is sometimes called a **CAM table**.

FIB

Name	Interface
/hufs.ac.kr /videos	1
/hufs.ac.kr /audio	2